

# Modelling Financial Markets

## Seminar 3: Factor Models and Asset Pricing — Hands-on




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# Acronyms used in this material

- **AI**: Artificial Intelligence
- **BAB**: Betting Against Beta
- **BET**: Bucharest Exchange Trading (index)
- **BH**: Benjamini–Hochberg (procedure)
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CAPM**: Capital Asset Pricing Model
- **CMA**: Conservative Minus Aggressive (investment factor)
- **CML**: Capital Market Line
- **ETF**: Exchange-Traded Fund
- **FDR**: False Discovery Rate
- **FF3**: Fama–French three-factor model
- **FF5**: Fama–French five-factor model
- **FM**: Fama–MacBeth (regressions)
- **FWER**: Family-Wise Error Rate
- **GLS**: Generalised Least Squares
- **GMM**: Generalized Method of Moments
- **GRS**: Gibbons–Ross–Shanken (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HML**: High Minus Low (value factor)
- **KRS**: Kan–Robotti–Shanken (standard errors)
- **ML**: Maximum Likelihood
- **MOM**: Momentum (factor)
- **NW**: Newey–West (standard errors)
- **OLS**: Ordinary Least Squares
- **PC1**: first Principal Component
- **PC2**: second Principal Component
- **PC3**: third Principal Component
- **PCA**: Principal Component Analysis
- **RF**: Risk-Free rate
- **RMW**: Robust Minus Weak (profitability factor)
- **SMB**: Small Minus Big (size factor)
- **SML**: Security Market Line
- **US**: United States

## Today's Question and Seminar Structure

- Questions of the seminar
  - **does beta explain the differences in average returns across assets?**
  - **how precise are the estimates?**
- Outputs to produce
  - the derivations of the A exercises
  - one table and one figure per B exercise
  - the corrected C1 code
  - one interpretation sentence per exercise
- Notebook for today: open the seminar notebook in Colab
  - each task names its notebook section
- Exercises, types and order: the Exercise Map (next slide)
- Seminar: Prof. Daniel Traian Pele, danpele@ase.ro; Antoaneta Amza, PhD candidate, antoaneta.amza@csie.ase.ro

# Exercise Map

| Exercise | Question | Type | Model |
|---|---|---|---|
| A1 | how many of ten candidate factors are discoveries? | Solved | – |
| A2 | why does the tangency portfolio price every asset by beta? | Solved | – |
| B1 | how precisely are XLK's beta and alpha estimated? | Solved | – |
| B2 | do nine sector ETFs reject the CAPM? | Solved | A2, B1 |
| C1 | what is wrong in an AI momentum backtest? | Proposed | B1, primer |
| After the lecture | A3 [Solved], B3 [Solved], B4 [Proposed]: Vasicek beta, PCA and the number of factors, BVB betas | homework | B1, A3 |
| Extension packs | A2 Extended, A4–A8, B5–B11, C2, C3 (optional, after the seminar) | Solved or Proposed (marked on each) | named on each task |

- **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: estimation and inference on real data, each ending with an interpretation question; Part C: open questions and AI critique

# Data Used

| Series | Source, price used | Frequency | Sample |
|---|---|---|---|
| 9 sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY) | adjusted close | month-end | Jan 1999 – Jul 2026 |
| 11 sector ETFs (the 9 + XLRE, XLC), SPY | adjusted close | daily | 20 Jun 2018 – 18 Sep 2026 |
| Mkt-RF, SMB, HML, RMW, CMA, MOM, RF | Kenneth French Data Library, percent | monthly, daily | Jul 1963 – Jul 2026 |
| 25 size × B/M portfolios | Kenneth French Data Library, value-weighted | monthly | Jul 1963 – Jul 2026 |
| 11 BVB blue chips, BET | stocks: adjusted close; BET: close | daily | 6 Jan 2015 (or listing) – 18 Sep 2026 |

- Daily prices of the course data, one series per asset (field used: adjusted close for ETFs and stocks, close for BET)
- Multi-asset analyses: align the *prices* on common days first, then take returns; monthly returns from month-end prices
- Sectors: XLB materials, XLE energy, XLF financials, XLI industrials, XLK technology, XLP consumer staples, XLU utilities, XLV health care, XLY consumer discretionary, XLRE real estate, XLC communication; SPY: an ETF on the S&P 500 index
- BVB blue chips: TLV, SNP, BRD, TGN, FP, SNG, H2O, SNN, EL, DIGI, TEL

# Setup: From Prices to Excess Returns

```
# notebook section "Setup" defines
# prices(), factors(), monthly_excess()
ex, F = monthly_excess(['XLK.US',
                        'SPY.US'])
# month-end adjusted prices -> simple
# returns -> minus RF -> align with factors
print(len(ex), ex.index[0].date(),
      ex.index[-1].date())
print(ex['XLK.US'].head(3))
print(F[['Mkt-RF', 'RF']].head(3))
```

- Expected: $T = 331$ months, 1999-01-31 to 2026-07-31
- If the count differs, check the field, the month-end rule and the division by 100

| | Jan 1999 | Feb 1999 | Mar 1999 |
|---|---|---|---|
| $P_{t-\mathbf{1}}$, XLK | 12.1103 | 14.0359 | 12.6439 |
| $P_t$, XLK | 14.0359 | 12.6439 | 13.5835 |
| $R_t = P_t/P_{t-\mathbf{1}} - 1$ | 15.90% | −9.92% | 7.43% |
| RF (/100) | 0.35% | 0.35% | 0.43% |
| $R^e_t = R_t - \mathrm{RF}_t$ | 15.55% | −10.27% | 7.00% |
| Mkt-RF | 3.47% | −4.08% | 3.48% |

- Worked line: $14.0359/12.1103 - 1 = 15.90\%$; subtracting 0.35% gives 15.55%
- Adjusted prices include dividends; RF and Mkt-RF are already excess or rate series

# Notation Used in Every Exercise (1/2): Returns and Portfolios

- **Simple return** and **excess return** of asset $i$ in period $t$

$$R_{i,t} = \frac{P_{i,t}}{P_{i,t-\mathbf{1}}} - 1, \qquad R^e_{i,t} = R_{i,t} - R_{f,t}$$

  - $P_{i,t}$: the adjusted price (dividends reinvested); $R_{f,t}$: the risk-free rate of the period (RF)
  - $f_t$: a factor, an excess return itself (Mkt-RF unless stated); $\mathrm{R}^e_t$: the $N \times 1$ vector of the $N$ assets

- **Portfolio return**: weights $w_i$ applied to *simple* returns

$$R_{p,t} = \sum_{i=\mathbf{1}}^{N} w_i R_{i,t}$$

  - log returns $r_t = \ln(1 + R_t)$ add over *time*, simple returns add over *assets* (Chapter 0)
  - long–short: $+1/n$ on $n$ assets bought, $-1/n$ on $n$ assets sold short

- Monthly to annual: means $\times 12$, volatilities and Sharpe ratios $\times \sqrt{12}$
  - daily data: 252 instead of 12

# Notation Used in Every Exercise (2/2): Moments, Matrices, Cross-Sections

- Expected excess returns and covariance of the $N$ assets

$$\boldsymbol{\mu} = \mathbb{E}[\mathrm{R}_t^e] \ (N \times 1), \qquad \boldsymbol{\Sigma} = \mathrm{Cov}(\mathrm{R}_t^e) \ (N \times N)$$

  - $\boldsymbol{\Sigma}$ positive definite: every portfolio has positive variance, so $\boldsymbol{\Sigma}^{-1}$ exists
  - 1: the vector of ones; I: the identity matrix; $^\top$: transpose; $\mathrm{a}^\top \boldsymbol{\Sigma}^{-1}\mathrm{a}$: a quadratic form, a squared distance that accounts for correlations

- Maximum-likelihood (ML) moments divide by $T$

$$\hat{\boldsymbol{\Sigma}} = \frac{1}{T}\hat{\mathrm{E}}^\top\hat{\mathrm{E}}, \qquad \hat{\sigma}_f^2 = \frac{1}{T}\sum_{t=1}^{T}(f_t - \bar{f})^2$$

  - $\hat{\mathrm{E}}$: the $T \times N$ matrix of regression residuals; $\bar{f}$: the time-series mean of the factor; exception: the factor covariance of B8 divides by $T-1$

- **Cross-sectional** moments: across the $N$ assets at a given time, not over time
  - $\mathrm{Var}_{cs}(\hat{\beta}) = \frac{1}{N-1}\sum_i(\hat{\beta}_i - \bar{\beta})^2$ (the unbiased estimator), $\bar{\beta} = \frac{1}{N}\sum_i \hat{\beta}_i$; $\mathrm{Cov}_{cs}$ likewise
  - $\otimes$ (Kronecker product, B7): A $\otimes$ B replaces each element $a_{jk}$ of A by the block $a_{jk}$B

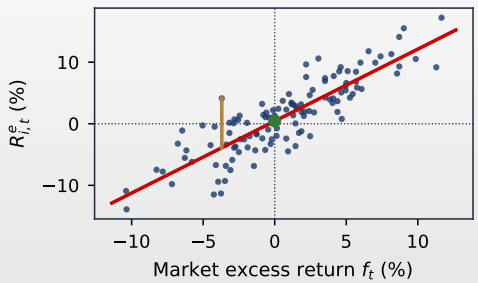
# What You Need for Today (1/21): Factor Models

- Details: Chapter 3; textbooks: Campbell, Lo & MacKinlay (1997) and, for principal components, Härdle & Simar (2019)
- **Factor model**: the excess return of an asset is explained by a few common factors

$$R^e_{i,t} = \alpha_i + \beta_{i,\mathbf{1}} f_{\mathbf{1},t} + \cdots + \beta_{i,K} f_{K,t} + \varepsilon_{i,t}$$

  - $f_{k,t}$: factor $k$, an excess return (a traded portfolio); $K$: the number of factors
  - $\beta_{i,k}$: the loading (exposure) of asset $i$ on factor $k$; $\varepsilon_{i,t}$: the idiosyncratic shock, mean 0
  - $\alpha_i$: the part of the mean return the factors do not explain; a correct model has $\alpha_i = 0$ for every asset
- Two questions of the seminar
  - do the betas explain why mean returns differ across assets (are the $\alpha_i$ zero)?
  - how precise are the estimates (standard errors robust to the features of returns)?

# What You Need for Today (2/21): The Market Model and OLS




- Months $(f_t, R^e_{i,t})$
- OLS: $\hat{\alpha} = 0.43$, $\hat{\beta} = 1.17$, $R^2 = 0.75$
- $\hat{\alpha}$: the line at $f_t = 0$
- one residual $\hat{\varepsilon}_t$

- **Market model**: one factor, $f_t$ = Mkt-RF

$$R^e_{i,t} = \alpha_i + \beta_i f_t + \varepsilon_{i,t}$$

- OLS (ordinary least squares) estimates, minimising $\sum_t \hat{\varepsilon}_t^2$

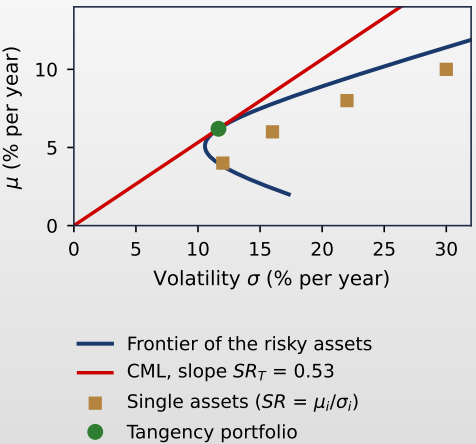
$$\hat{\beta}_i = \frac{\sum_t (f_t - \bar{f})(R^e_{i,t} - \bar{R}^e_i)}{\sum_t (f_t - \bar{f})^2}, \qquad \hat{\alpha}_i = \bar{R}^e_i - \hat{\beta}_i \bar{f}$$

- Reading
  - $\hat{\beta}_i$: the slope, $\widehat{\mathrm{Cov}}(R^e_i, f)/\widehat{\mathrm{Var}}(f)$; $\beta > 1$: more volatile than the market
  - $\hat{\alpha}_i$: the intercept, the mean excess return in a month with $f_t = 0$
  - $R^2 = 1 - \sum_t \hat{\varepsilon}_t^2 / \sum_t (R^e_{i,t} - \bar{R}^e_i)^2$: the share of the variance explained by the market

# What You Need for Today (3/21): The Fama–French Factors

- Factors are long–short portfolios sorted on a characteristic (Fama & French (1993); Fama & French (2015); Carhart (1997)); data: Kenneth French Data Library
  - **Mkt-RF**: the value-weighted US market minus the risk-free rate
  - **SMB** (Small Minus Big): small minus large firms; **HML** (High Minus Low): high minus low book-to-market (value minus growth)
  - **RMW** (Robust Minus Weak): high minus low operating profitability; **CMA** (Conservative Minus Aggressive): low minus high asset growth
  - **MOM**: past winners minus past losers (12–1 momentum, What You Need (11/21))
- The models
  - CAPM: Mkt-RF; FF3: Mkt-RF, SMB, HML; Carhart: FF3 + MOM; FF5: Mkt-RF, SMB, HML, RMW, CMA
  - FF3 regression: $R^e_{i,t} = \alpha_i + b_i\,\text{Mkt-RF}_t + s_i\,\text{SMB}_t + h_i\,\text{HML}_t + \varepsilon_{i,t}$
- Test assets
  - 25 size $\times$ B/M portfolios: stocks sorted into 5 size groups $\times$ 5 book-to-market groups
  - ETF (exchange-traded fund): a fund listed on an exchange; sector ETFs hold one sector, factor ETFs follow a factor (MTUM: momentum, QUAL: quality)

# What You Need for Today (4/21): Mean–Variance and the Tangency Portfolio

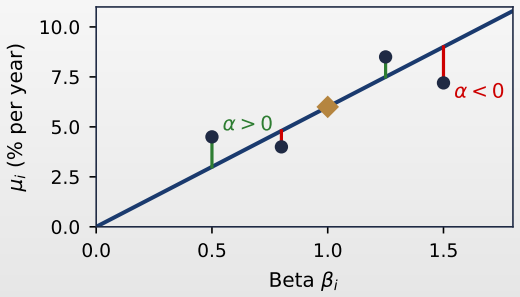



- **Sharpe ratio**: mean excess return per unit of volatility, $SR = \mu/\sigma$
  - mean–variance investors (Markowitz (1952)) with a risk-free asset all hold the risky portfolio with the highest $SR$: the **tangency portfolio**
- Weights and squared Sharpe ratio of the tangency portfolio

$$w_T = \frac{\boldsymbol{\Sigma^{-1}\mu}}{1^\top \boldsymbol{\Sigma^{-1}\mu}}, \qquad SR_T^2 = \boldsymbol{\mu}^\top \boldsymbol{\Sigma^{-1}\mu}$$

- **CML** (capital market line): the risk-free asset mixed with the tangency portfolio
  - chart: four assets, $SR$ from 0.33 to 0.38; $SR_T = 0.53$
  - example: two uncorrelated assets, $\boldsymbol{\mu} = (4\%, 6\%)$, variances 0.04 and 0.09: $w_T = (0.60, 0.40)$, $SR_T^2 = 0.04 + 0.04$, $SR_T = 0.283$

# What You Need for Today (5/21): The CAPM and the Security Market Line

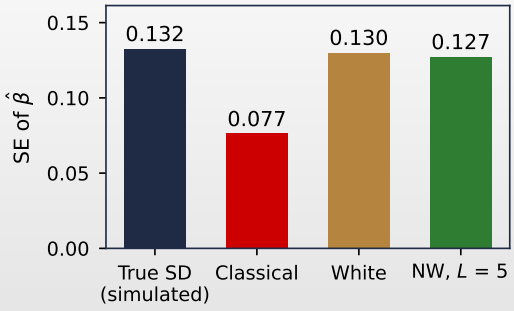



- **CAPM** (Sharpe (1964)): if everyone holds the tangency portfolio, it is the market, and for every asset

$$\mu_i = \beta_i \mu_m, \qquad \beta_i = \frac{\mathrm{Cov}(R_i^e, R_m^e)}{\mathrm{Var}(R_m^e)}$$

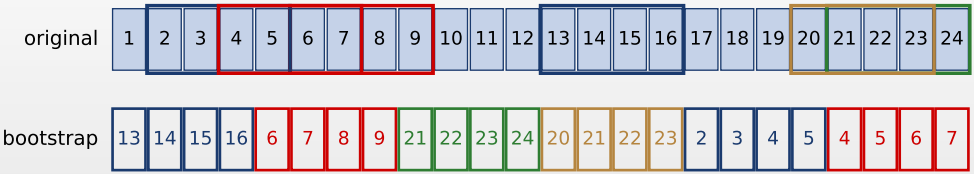
  - $\mu_i = \mathbb{E}[R_i^e]$, $\mu_m = \mathbb{E}[R_m^e]$: expected excess returns of the asset and of the market
  - SML (security market line): only market risk is priced; $\varepsilon_{i,t}$ is diversified away
- **Alpha**: the vertical distance to the SML
  - testable consequence: $\alpha_i = 0$ for every asset
  - example: $\beta = 1.2$, premium 6% a year: CAPM gives 7.2%; a realised 9% gives $\alpha = 1.8\%$
- A rejection may mean that the market proxy is not efficient (A2)
  - zero-beta CAPM (Black (1972)): $\mu_i - \gamma = \beta_i(\mu_m - \gamma)$, $\gamma$: the mean of a zero-beta portfolio

# What You Need for Today (6/21): Classical, White and Newey–West Standard Errors



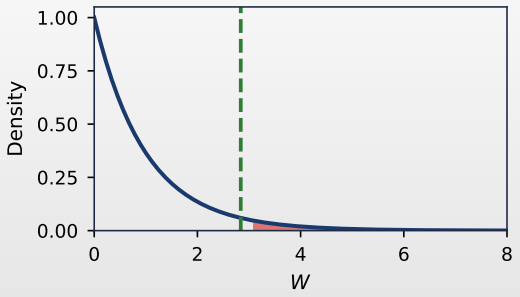

- OLS of $\hat{\boldsymbol{\theta}} = (\hat{\alpha}, \hat{\beta})^\top$ on $x_t = (1, f_t)^\top$; X: rows $x_t^\top$; $\hat{\varepsilon}_t$: residuals
$$\widehat{\mathrm{Var}}(\hat{\boldsymbol{\theta}}) = (\mathrm{X}^\top \mathrm{X})^{-1} \hat{\mathrm{S}} (\mathrm{X}^\top \mathrm{X})^{-1}$$
- Choices of $\hat{\mathrm{S}}$
  - classical: $\hat{\sigma}_\varepsilon^2 \mathrm{X}^\top \mathrm{X}$ (constant variance, no autocorrelation)
  - White (White (1980)): $\sum_t \hat{\varepsilon}_t^2 x_t x_t^\top$, robust to heteroskedasticity
  - Newey–West (Newey & West (1987)), a HAC estimator: adds $\sum_{l=1}^{L} \left(1 - \frac{l}{L+1}\right) \sum_t \hat{\varepsilon}_t \hat{\varepsilon}_{t-l} (x_t x_{t-l}^\top + x_{t-l} x_t^\top)$
  - lag rule $L = \lfloor 4(T/100)^{2/9} \rfloor$; $T = 331$: $L = 5$
- Chart: 2 000 simulated samples with autocorrelated residuals that are larger in volatile months
  - the classical standard error (0.077) understates the true standard deviation of $\hat{\beta}$ (0.132); White and NW are close to it

# What You Need for Today (7/21): Bootstrap Intervals




- **Moving-block bootstrap** (Künsch (1989)): draw blocks of $b$ consecutive months (here $b = 4$) and concatenate them to length $T$
  - resample the pairs $(R^e_t, f_t)$ together; re-estimate; repeat $B$ times; the 2.5% and 97.5% quantiles form a 95% percentile interval
  - the dependence inside each block (autocorrelation, volatility clustering) is kept; block starts may overlap
- Variants
  - **stationary bootstrap** (Politis & Romano (1994)): random block lengths, geometric with mean $b$; **i.i.d.** bootstrap (independent and identically distributed draws): single months ($b = 1$)
  - bootstrap *under* $H_0$: rebuild data with the null imposed; $p = (1 + \#\{W^* \geq W\})/(B + 1)$

# What You Need for Today (8/21): The GRS Test




- $F(2, 118)$ density under $H_0$
- rejection region, 5%: $W > 3.07$
- example $W = 2.84$, $p = 0.062$

- **GRS** (Gibbons, Ross & Shanken (1989)): $H_0$: $\alpha_1 = \cdots = \alpha_N = 0$ jointly

$$W = \frac{T - N - 1}{N} \cdot \frac{\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \hat{\mu}_m^2 / \hat{\sigma}_m^2} \sim F(N, T - N - 1)$$

- Symbols
  - $N$: number of test assets; $T$: number of months; $\hat{\alpha}$: the $N$ alphas; $\hat{\Sigma}$: ML residual covariance
  - $\hat{\mu}_m / \hat{\sigma}_m$: the factor's Sharpe ratio (ML); $F$: the Fisher distribution, exact for i.i.d. Normal errors
- Reading
  - large $W$: the alphas are far from 0 relative to their noise; reject when $W$ exceeds the 95% quantile of $F$
  - example: $T = 121$, $N = 2$, $\hat{\alpha}^\top \hat{\Sigma}^{-1} \hat{\alpha} = 0.05$, $SR_m = 0.2$: $W = 59 \times 0.05 / 1.04 = 2.84 < 3.07$, $p = 0.062$
  - robust alternative (B7): a Wald test $\hat{\alpha}^\top \hat{V}_\alpha^{-1} \hat{\alpha} \to \chi^2_N$ with a sandwich covariance $\hat{V}_\alpha$

# What You Need for Today (9/21): Normality of the Residuals

- The $F$ distribution of GRS assumes residuals with the Normal distribution (B7)
  - heavy tails change the distribution of $W$; the bootstrap under $H_0$ (What You Need (7/21)) does not need normality
- **Pearson kurtosis** of a residual series

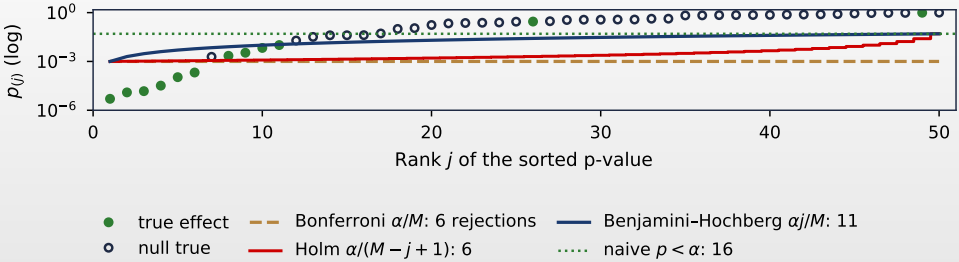
$$\text{Kurt} = \frac{\frac{1}{T}\sum_t \hat{\varepsilon}_t^4}{\hat{\sigma}^4}, \qquad \hat{\sigma}^2 = \frac{1}{T}\sum_t \hat{\varepsilon}_t^2$$

  - Kurt $= 3$ for the Normal distribution; Kurt $> 3$: heavy tails, extreme months more frequent
- **Q–Q plot**: sorted standardised residuals against Normal quantiles
  - points on the 45-degree line: Normal; an S shape: heavy tails (Seminar 1)
- **Wald test**, general form: a vector of estimates $\hat{\boldsymbol{\theta}}$ with covariance $\hat{\mathrm{V}}$

$$W = \hat{\boldsymbol{\theta}}^\top \hat{\mathrm{V}}^{-1} \hat{\boldsymbol{\theta}} \to \chi^2_q \text{ under } H_0 : \boldsymbol{\theta} = 0$$

  - $q$: the number of restrictions; $\chi^2_q$: the chi-squared distribution with $q$ degrees of freedom

# What You Need for Today (10/21): Multiple Testing, FWER and FDR




- With $M$ tests some p-values are small by chance (hundreds of published factors, Harvey, Liu & Zhu (2016))
  - **FWER**: $P$(at least one false rejection); Bonferroni $p_i \leq \alpha/M$; Holm (Holm (1979)): $p_{(j)} \leq \alpha/(M-j+1)$ until the first failure
  - **FDR** (false discovery rate): expected share of false discoveries among rejections; **Benjamini–Hochberg** (Benjamini & Hochberg (1995)): largest $j$ with $p_{(j)} \leq \alpha j/M$, reject $H_{(1)}, \ldots, H_{(j)}$
  - BH: independent or positively dependent tests; Benjamini–Yekutieli (Benjamini & Yekutieli (2001)): the BH thresholds divided by $\sum_{i=1}^{M} 1/i$, any dependence
  - chart: 12 true effects among 50 tests; BH reports 11 discoveries (one false), Holm 6

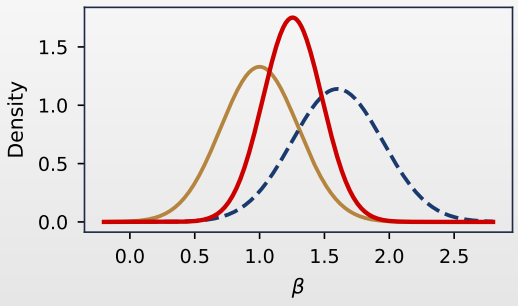
# What You Need for Today (11/21): Momentum and Look-Ahead Bias

- **Momentum** (Jegadeesh & Titman (1993)): rank stocks on the past return, buy the winners, sell the losers
  - 12–1 signal: the compound return of months $t-12, \ldots, t-2$, known at the end of $t-1$; month $t-1$ is skipped

$$s_{i,t} = \prod_{k=2}^{12} (1 + R_{i,t-k}) - 1$$

  - the portfolio is held during month $t$; its return averages the simple returns of month $t$ (Notation (1/2))
- **Look-ahead bias**: the signal uses information not yet available at formation
  - test: the weights must not change if any price after the end of $t-1$ changes
- Other pitfalls of a backtest (C2, C1)
  - **survivorship bias**: today's blue chips are chosen with hindsight; losers that left the index are missing
  - short-sale restrictions, costs and liquidity reduce what can be earned; with 3 stocks per side, one stock can drive the result of a month

# What You Need for Today (12/21): Bayesian Shrinkage of Beta

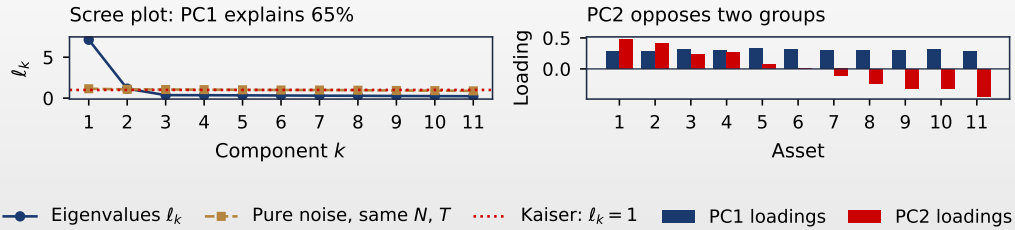



- Prior $\beta_i \sim N(\bar{\beta}, \sigma_\beta^2)$; estimate $\hat{\beta}_i \mid \beta_i \sim N(\beta_i, \text{se}_i^2)$
  - $\bar{\beta}$, $\sigma_\beta^2$: mean and true dispersion of the betas across assets; $\text{se}_i$: the standard error of $\hat{\beta}_i$
- **Vasicek** (Vasicek (1973)): the posterior mean shrinks $\hat{\beta}_i$ toward $\bar{\beta}$

$$\beta_i^V = w_i \hat{\beta}_i + (1 - w_i)\bar{\beta}, \qquad w_i = \frac{\sigma_\beta^2}{\sigma_\beta^2 + \text{se}_i^2}$$

- Reading and estimation
  - noisy $\hat{\beta}_i$ (large $\text{se}_i$): small $w_i$, strong shrinkage; chart: $w = 0.42$
  - law of total variance: $\text{Var}(\hat{\beta}) = \text{Var}(\mathbb{E}[\hat{\beta} \mid \beta]) + \mathbb{E}[\text{Var}(\hat{\beta} \mid \beta)]$, used to estimate $\sigma_\beta^2$ (A3)
  - **Blume** (Blume (1971)): betas regress toward 1; the practitioners' rule $0.33 + 0.67\hat{\beta}_i$ is a fixed pull toward 1

# What You Need for Today (13/21): Principal Components




- Standardise each return series (Z: $T \times N$), then decompose the correlation matrix

$$\mathrm{C} = \tfrac{1}{T}\mathrm{Z}^\top \mathrm{Z} = \sum_{k=1}^{N} \ell_k \boldsymbol{\gamma}_k \boldsymbol{\gamma}_k^\top, \qquad \ell_1 \geq \cdots \geq \ell_N, \quad \textstyle\sum_k \ell_k = N$$

- $\ell_k$: eigenvalue (written $\ell$, not $\lambda$, which is the risk premium), the variance of component $k$; share $\ell_k / N$; $\boldsymbol{\gamma}_k$: eigenvector of norm 1 (loadings); scores $\mathrm{Z}\boldsymbol{\gamma}_k$
  - chart: 11 simulated assets, a market factor and a weaker one; PC1: all loadings positive; PC2 opposes two groups

# What You Need for Today (14/21): How Many Factors?

- **Kaiser** (Kaiser (1960)): keep the components with $\ell_k > 1$ (more variance than one standardised asset)
- **Bai–Ng** (Bai & Ng (2002)): penalised information criterion

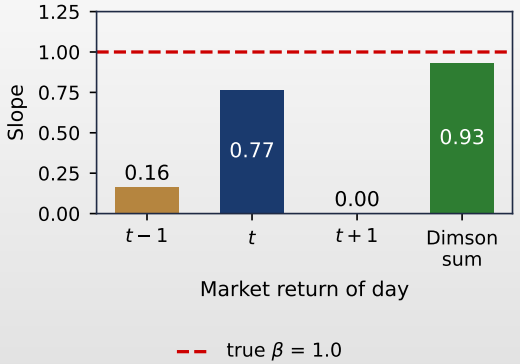
$$IC_{p\mathbf{2}}(k) = \ln V(k) + k\,\frac{N+T}{NT}\ln\min(N,T), \qquad V(k) = \frac{1}{N}\sum_{j>k}\ell_j$$

  - $V(k)$: the variance left after $k$ factors; choose $\hat{k} = \arg\min_{\mathbf{0}\le k\le k_{\max}} IC_{p\mathbf{2}}(k)$
  - designed for large $N$ and $T$; with $N = 11$ the penalty is weak and $\hat{k}$ tends to $k_{\max}$
- **Ahn–Horenstein** (Ahn & Horenstein (2013)): eigenvalue ratio

$$\hat{k} = \arg\max_{\mathbf{1}\le k\le k_{\max}} \frac{\ell_k}{\ell_{k+\mathbf{1}}}$$

  - the largest drop in the scree plot; in the chart of What You Need (13/21), at $k = 1$
- Pure noise (amber in the chart of What You Need (13/21)) has all eigenvalues near 1: compare with it before naming a factor

# What You Need for Today (15/21): Stale Prices and the Dimson Beta




- **Stale prices**: an illiquid stock does not trade every day, so its price reacts to market news with a delay
  - the same-day OLS beta is biased toward 0
- **Dimson beta** (Dimson (1979)): add the lagged and the leading market return and sum the slopes

$$r_{i,t} = a + b_{-1} r_{m,t-1} + b_0 r_{m,t} + b_{+1} r_{m,t+1} + e_t$$

$$\beta^D = b_{-1} + b_0 + b_{+1}$$

- Chart (simulated, true beta 1, the stock trades on 75% of days)
  - same-day slope 0.77; the lag adds 0.16; Dimson 0.93, close to 1
  - the lead and lag must be formed on the full calendar, before any day is removed

# What You Need for Today (16/21): Two-Pass (Fama–MacBeth) Regressions

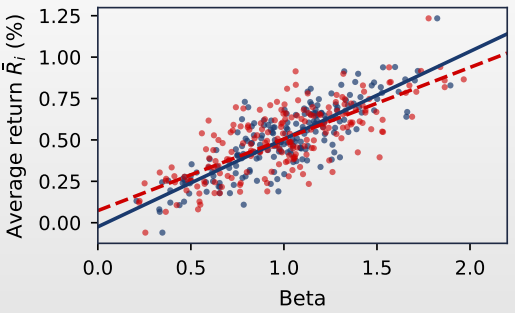
- **Pass 1** (time series): for each asset, estimate the betas $\hat{\boldsymbol{\beta}}_i$ ($K \times 1$) by OLS; $\hat{\mathrm{B}}$: $N \times K$
- **Pass 2** (Fama & MacBeth (1973); cross-section, every month $t$): regress the $N$ returns on the betas

$$R^e_{i,t} = \lambda_{\mathbf{0},t} + \hat{\boldsymbol{\beta}}_i^\top \boldsymbol{\lambda}_t + e_{i,t}, \qquad \hat{\boldsymbol{\lambda}} = \frac{1}{T}\sum_{t=\mathbf{1}}^{T} \hat{\boldsymbol{\lambda}}_t$$

  - $\boldsymbol{\lambda}$: the risk premia (prices of risk) of the factors; $\lambda_{\mathbf{0}}$: the zero-beta rate, 0 if the model is right
  - for a traded factor the model also implies $\lambda = \mathbb{E}[f]$

- Standard errors
  - Fama–MacBeth: $\mathrm{se}(\hat{\lambda}) = \mathrm{sd}(\hat{\lambda}_t)/\sqrt{T}$; Newey–West on the series $\hat{\lambda}_t$ if it is autocorrelated
  - Shanken correction (Shanken (1992)): the betas are estimated, so $\mathrm{Var}_{\mathbf{Sh}} = (1+c)(\mathrm{Var}_{FM} - \boldsymbol{\Sigma}_f/T) + \boldsymbol{\Sigma}_f/T$, $c = \hat{\boldsymbol{\lambda}}^\top \boldsymbol{\Sigma}_f^{-\mathbf{1}} \hat{\boldsymbol{\lambda}}$
  - $\boldsymbol{\Sigma}_f$: covariance of the factors; Kan–Robotti–Shanken (Kan, Robotti & Shanken (2013)): a GMM version that also allows misspecified models

# What You Need for Today (17/21): Errors in Variables and Useless Factors




- true $\beta_i$: slope 0.53
- estimated $\hat{\beta}_i$: slope 0.43

- **Errors in variables**: pass 2 uses $\hat{\beta}_i = \beta_i + \eta_i$, with estimation noise $\eta_i$
  - the slope on a noisy regressor is attenuated toward 0 by the factor $\text{Var}_{cs}(\beta)/[\text{Var}_{cs}(\beta) + \text{Var}(\eta)]$
  - chart: 200 simulated assets, $T = 60$; true slope 0.53, on estimated betas 0.43
  - remedies: portfolios (smaller $\text{Var}(\eta)$), long samples, the Shanken correction for the standard error
- **Useless factor** (Kan & Zhang (1999)): a factor unrelated to returns has true betas equal to 0
  - its estimated betas are pure noise, yet pass 2 can report a large, "significant" premium (A7)
  - check first that the betas differ from 0 jointly (pass 1)

# What You Need for Today (18/21): Cross-Sectional Fit, OLS and GLS

- Fit of mean returns $\boldsymbol{\mu}$ ($N \times 1$) by $\mathsf{X} = [1, \hat{\mathsf{B}}]$

$$R^2_{OLS} = 1 - \frac{\mathsf{e}^\top \mathsf{e}}{\sum_i (\mu_i - \bar{\mu})^2}, \qquad \mathsf{e} = \boldsymbol{\mu} - \mathsf{X}\hat{\boldsymbol{\lambda}}^+$$

  - $\hat{\boldsymbol{\lambda}}^+ = (\mathsf{X}^\top \mathsf{X})^{-1} \mathsf{X}^\top \boldsymbol{\mu}$: the intercept $\hat{\lambda}_0$ and the premia $\hat{\boldsymbol{\lambda}}$; $\bar{\mu}$: the average of the $\mu_i$
  - with few, strongly factor-driven test assets (25 size $\times$ B/M) almost any model with the right factors reaches a high $R^2_{OLS}$ (Lewellen, Nagel & Shanken (2010))

- **GLS**: weight the pricing errors by the inverse covariance $\hat{\mathsf{V}}^{-1}$ of the returns

$$R^2_{GLS} = 1 - \frac{\mathsf{e}_G^\top \hat{\mathsf{V}}^{-1} \mathsf{e}_G}{(\boldsymbol{\mu} - \bar{\mu}_G 1)^\top \hat{\mathsf{V}}^{-1} (\boldsymbol{\mu} - \bar{\mu}_G 1)}$$

  - it measures the fit of the portfolios that matter for the maximum Sharpe ratio, not only of the 25 averages
  - $\mathsf{e}_G = \boldsymbol{\mu} - \mathsf{X}\hat{\boldsymbol{\lambda}}^+_{GLS}$, $\hat{\boldsymbol{\lambda}}^+_{GLS} = (\mathsf{X}^\top \hat{\mathsf{V}}^{-1} \mathsf{X})^{-1} \mathsf{X}^\top \hat{\mathsf{V}}^{-1} \boldsymbol{\mu}$; $\bar{\mu}_G = 1^\top \hat{\mathsf{V}}^{-1} \boldsymbol{\mu} / 1^\top \hat{\mathsf{V}}^{-1} 1$: the GLS mean
  - remedy of Lewellen, Nagel & Shanken (2010): add test assets with other characteristics (industries) and report both $R^2$

# What You Need for Today (19/21): Comparing Factor Models by Sharpe Ratios

- Maximum squared Sharpe ratio attainable with the $K$ factors of a model (Barillas & Shanken (2018))

$$SR^2 = \boldsymbol{\mu}_f^\top \boldsymbol{\Sigma}_f^{-1} \boldsymbol{\mu}_f$$

  - $\boldsymbol{\mu}_f$, $\boldsymbol{\Sigma}_f$: means and covariance of the factors; the better model prices more assets and has the higher $SR^2$
- **Nested** models (one contains the other): a **spanning regression**

$$f_{\mathbf{new},t} = a + \mathrm{b}^\top \mathrm{f}_t + u_t$$

  - $a \neq 0$: the new factor raises the attainable Sharpe ratio; $a = 0$: it is spanned (redundant); test $a$ with Newey–West
- **Non-nested** models (Barillas, Kan, Robotti & Shanken (2020)): bootstrap the difference $SR_A^2 - SR_B^2$
  - use the same resampled months for both models, so that the difference keeps their correlation

# What You Need for Today (20/21): Rolling and Historical Betas

- **Rolling beta**: estimate $\beta_{i,t}$ on the last $H$ months up to $t-1$ only (C2: $H = 36$, at least 18)
  - a beta-sorted portfolio formed at $t-1$ is then implementable, without look-ahead
- **Low-beta anomaly** (Frazzini & Pedersen (2014)): low-beta stocks earn more than the SML predicts, high-beta stocks less
  - BAB (betting against beta): leverage the low-beta side and scale the high-beta side so that the portfolio beta is 0
  - an unscaled low-minus-high spread has a negative beta: part of its return is market exposure
- **Regression to the mean**: betas estimated in one period move toward 1 in the next
  - regress later betas on earlier ones across assets: a slope below 1 measures the pull toward the mean (A5)

# What You Need for Today (21/21): Tools of the Replication Project (Pack 8)

- **Factor themes**: hierarchical clustering of the factors (Ward's method, Ward (1963))
  - distance $d_{jk} = 1 - \text{corr}(\hat{\varepsilon}_j, \hat{\varepsilon}_k)$ between CAPM residuals; Ward merges the two clusters whose union raises the within-cluster variance least
  - the dendrogram shows the merges; cutting it at a height gives the themes
- **Hierarchical Bayes** for many alphas (Jensen, Kelly & Pedersen (2023))

$$\hat{\boldsymbol{\alpha}} \sim N\left(0,\ \mathrm{M}\mathrm{M}^{\top}\tau_c^2 + \mathrm{I}\tau_w^2 + \hat{\boldsymbol{\Sigma}}/T\right)$$

  - M: membership of factors in themes; $\tau_c$: dispersion of theme alphas; $\tau_w$: of factor alphas within a theme; $\hat{\boldsymbol{\Sigma}}/T$: sampling noise
  - the posterior mean of each alpha borrows strength from its theme, as Vasicek borrows from the cross-section (What You Need (12/21))
- **Post-publication decay**: regress the out-of-sample alphas on the in-sample ones
  - a slope below 1: part of the published alpha was noise or was arbitraged away

# A1 [Solved]: Ten Candidate Factors — Task

- **Question**: how many of ten new factors are discoveries once we account for testing ten at a time?
- **Inputs**: a new example (not the vector of Chapter 3); level $\alpha = 5\%$, $M = 10$ two-sided tests of zero mean
- **Tasks**: complete the worksheet
  1. Count the naive discoveries $p \leq 0.05$ and the Bonferroni discoveries $p \leq 0.005$.
  2. Apply Holm: compare $p_{(j)}$ with $0.05/(M - j + 1)$ and stop at the first failure.
  3. Apply Benjamini–Hochberg (BH): find the largest $j$ with $p_{(j)} \leq 0.005j$ and reject $H_{(1)}, \ldots, H_{(j)}$.
  4. State which error rate each procedure controls and under which dependence; as an extension, also apply Benjamini–Yekutieli (BY).

| $j$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $p_{(j)}$ | 0.001 | 0.004 | 0.009 | 0.012 | 0.021 | 0.035 | 0.048 | 0.060 | 0.20 | 0.55 |
| Holm $0.05/(M - j + 1)$ | 0.0050 | ... | | | | | | | | |
| BH $0.005j$ | 0.005 | ... | | | | | | | | |

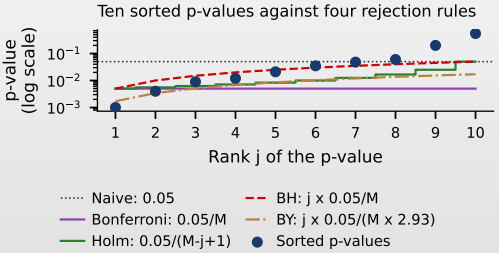
- **Report**: the completed worksheet and the four counts of discoveries
- Notebook: section A1

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# A1 [Solved]: Solution

| $j$ | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| $p_{(j)}$ | 0.001 | 0.004 | 0.009 | 0.012 | 0.021 | 0.035 | 0.048 | 0.060 | 0.20 | 0.55 |
| Holm | **0.0050** | **0.0056** | 0.0063 | 0.0071 | 0.0083 | 0.0100 | 0.0125 | 0.0167 | 0.025 | 0.05 |
| BH | **0.005** | **0.010** | **0.015** | **0.020** | **0.025** | 0.030 | 0.035 | 0.040 | 0.045 | 0.050 |

- Naive **7**; Bonferroni **2**; Holm: $0.009 > 0.0063$ is the first failure, so **2**
- BH: $j = 5$ is the largest rank with $p_{(j)} \leq 0.005j$ ($0.021 \leq 0.025$), so **5**
  - for $j = 6, \ldots, 10$: $0.035 > 0.030$; $0.048 > 0.035$; $0.060 > 0.040$; $0.20 > 0.045$; $0.55 > 0.050$
- Error rate controlled
  - Bonferroni and Holm: FWER, under any dependence
  - BH: FDR, under independence or positive dependence
  - BY (the BH thresholds divided by $\sum_{i=1}^{10} 1/i = 2.93$): FDR, under any dependence; **1** discovery
- Interpretation: a factor search tolerates a known share of false factors, so FDR is the relevant target



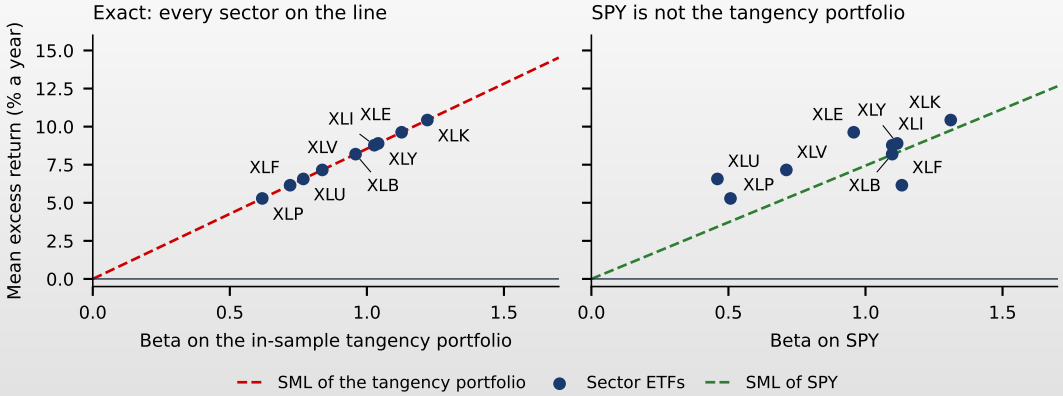

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# A2 [Solved]: The CAPM from the Tangency Portfolio — Task

- **Question**: if the market is the tangency portfolio, why must every asset satisfy $\mu_i = \beta_i \mu_m$?
- **Inputs**: $N$ risky assets with expected excess returns $\boldsymbol{\mu}$ ($N \times 1$) and positive definite covariance $\boldsymbol{\Sigma}$; a risk-free asset
- **Assumption**: the market is the tangency portfolio $\mathrm{w}_m = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}/c$, with the scalar $c = 1^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} \neq 0$; $R^e_m = \mathrm{w}_m^\top \mathrm{R}^e$
- **Tasks**
  1. Express the covariance vector $\mathrm{Cov}(\mathrm{R}^e, R^e_m) = \boldsymbol{\Sigma}\mathrm{w}_m$ as a function of $\boldsymbol{\mu}$ and $c$.
  2. Compute $\mu_m = \mathrm{w}_m^\top \boldsymbol{\mu}$ and $\sigma^2_m = \mathrm{w}_m^\top \boldsymbol{\Sigma} \mathrm{w}_m$ and find the relation between them.
  3. Combine steps 1 and 2 to prove that $\mu_i = \beta_i \mu_m$ for every asset, with $\beta_i = \mathrm{Cov}(R^e_i, R^e_m)/\sigma^2_m$.
  4. Check the result on the 9 sector ETFs (monthly, 1999–2026): replace $\boldsymbol{\mu}$, $\boldsymbol{\Sigma}$ by sample moments and compare $\hat{\mu}_i$ with $\hat{\beta}_i \hat{\mu}_m$, first for the sample tangency portfolio, then for SPY.
- **Report**: the derivation in three steps, a chart of the two checks and one sentence
- Notebook: section A2

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# A2 [Solved]: Solution (1/2) — Chart




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# A2 [Solved]: Solution (2/2) — Steps of the Proof

1. $\boldsymbol{\Sigma} \mathrm{w}_m = \boldsymbol{\mu}/c$, so $\mathrm{Cov}(\mathrm{R}^e, R_m^e) = \boldsymbol{\mu}/c$
2. $\sigma_m^2 = \mathrm{w}_m^\top \boldsymbol{\Sigma} \mathrm{w}_m = \mu_m/c$, hence $c = \mu_m/\sigma_m^2$
3. $\boldsymbol{\mu} = c\ \mathrm{Cov}(\mathrm{R}^e, R_m^e) = \boldsymbol{\beta}\,\mu_m$: the SML
4. Sample tangency portfolio (mean 8.55% a year): largest gap $\approx 3 \times 10^{-17}$; SPY: gaps up to 3.1% a year (XLU +3.1%, XLE +2.5%, XLF −2.3%)

⊡ Any efficient portfolio prices all assets exactly; a CAPM test asks whether the market proxy is efficient

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# B1 [Solved]: Beta and Alpha of XLK — Task

- **Questions**
  - what are the CAPM beta and alpha of the technology ETF?
  - how precisely are they known?
- **Data**: XLK and Mkt-RF, monthly excess returns, January 1999 – July 2026 ($T = 331$), as in the Setup slide
- **Tasks**
  1. Estimate $R^e_{\text{XLK},t} = \alpha + \beta\,\text{Mkt-RF}_t + \varepsilon_t$ by OLS; annualise $\alpha$ and its standard errors by multiplying by 12.
  2. Compute classical, White (heteroskedasticity-robust) and Newey–West standard errors (s.e.) of $\alpha$ and $\beta$, with $L = 5$ lags.
  3. Test $H_0 : \alpha = 0$ at 5% with the Newey–West $t$-statistic.
  4. Apply a moving-block bootstrap (Künsch (1989)): resample blocks of 12 consecutive months of the pairs $(R^e_t, f_t)$ and re-estimate $\alpha$ in 2000 draws, with seed 42; report the 95% percentile interval.
  5. Plot the data with the fitted line, the residuals over time and the bootstrap distribution.
- **Report**: a table (estimate, three standard errors), $t(\alpha)$, $R^2$, the interval and the chart
- **Interpretation**: the HAC error of beta is twice the classical one: which feature of the residuals causes it?
- Notebook: section B1

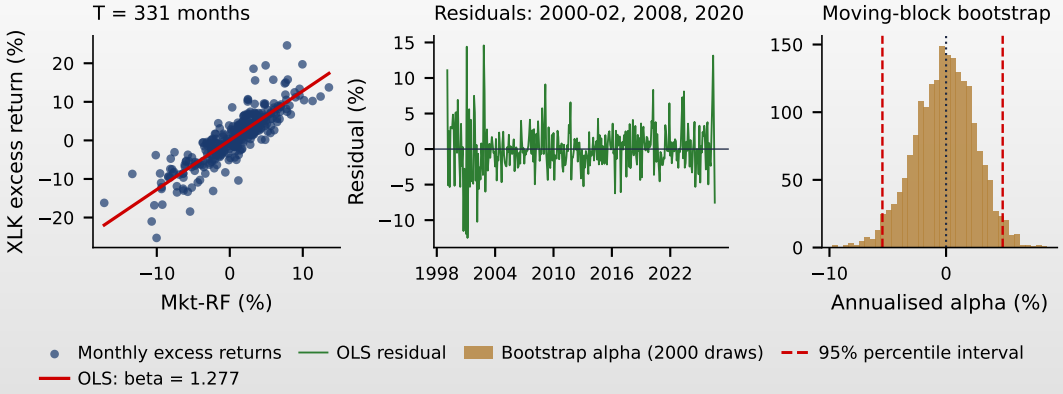
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# B1 [Solved]: Solution (1/3) — Step by Step

| | Estimate | Classical s.e. | White s.e. | Newey–West s.e. |
|---|---|---|---|---|
| $\alpha$ (annualised) | 0.33% | 2.27% | 2.32% | 2.45% |
| $\beta$ | 1.277 | 0.041 | 0.054 | 0.082 |

1. $R^2 = 0.742$; monthly $\hat{\alpha} = 0.027\%$, so $12 \times 0.027\% = 0.33\%$ a year
2. Lags $L = \lfloor 4 \times 3.31^{2/9} \rfloor = \lfloor 5.2 \rfloor = 5$; robust 95% interval of $\beta$: $1.277 \pm 1.96 \times 0.082 = [1.12, 1.44]$ instead of $[1.20, 1.36]$
3. $t(\alpha) = 0.33/2.45 = 0.13$, $p = 0.89$: do not reject $H_0$; no evidence of a non-zero CAPM alpha
4. Bootstrap: $\alpha \in [-5.5\%, 4.9\%]$, standard deviation of the draws 2.61%; the interval contains 0

# B1 [Solved]: Solution (2/3) — Chart




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# B1 [Solved]: Solution (3/3) — Interpretation

- Classical errors assume $\mathrm{Var}(\varepsilon_t \mid f_t)$ constant and no serial correlation
  - the sandwich estimator weights each squared residual by $(f_t - \bar{f})^2$
- White raises the s.e. of $\beta$ from 0.041 to 0.054: heteroskedasticity
  - residuals are large when the market moves a lot (2000–2002, 2008, 2020)
- Newey–West raises it further to 0.082: serial correlation
  - the products $f_t \varepsilon_t$ are serially correlated, so the cross terms of the HAC estimator are positive
- Answer to the interpretation question: the doubling of the s.e. of $\beta$ comes from the heteroskedasticity and the serial correlation of the residuals
- The point estimates are the same; only the uncertainty changes

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# B2 [Solved]: GRS Test on Sector ETFs — Task

- **Question**: do the nine sector ETFs jointly reject the CAPM?
- **Data**: 9 sector ETFs, month-end adjusted prices aligned on common months; monthly excess returns, January 1999 – July 2026 ($T = 331$, $N = 9$); factor Mkt-RF
- **Tasks**
  1. Estimate the 9 time-series regressions on Mkt-RF; report the annual alphas with Newey–West intervals and the betas.
  2. Build the residual matrix $\hat{\mathrm{E}}$ ($T \times 9$), $\hat{\boldsymbol{\Sigma}} = \hat{\mathrm{E}}^\top \hat{\mathrm{E}}/T$ and the quadratic form $\hat{\boldsymbol{\alpha}}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\alpha}}$ with monthly alphas.
  3. Compute $\hat{\mu}_m$, $\hat{\sigma}_m$ (ML), then the GRS statistic $W$ (What You Need (8/21)) and its $p$-value from $F(9, 321)$.
  4. Repeat the test on the 25 size × B/M portfolios over the same 331 months.
- **Report**: the coefficient table, the three inputs of $W$, $W$, $p$, the chart and the same-period comparison
- **Interpretation**: the 25 portfolios reject the CAPM (GRS = 4.20 over 1963–2026); why do the sectors not reject it?
- Model: A2 (efficiency), B1 (regression); Notebook: section B2

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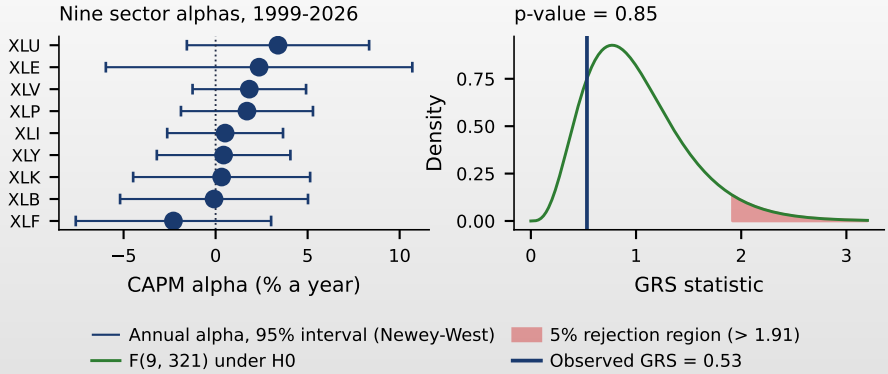
# B2 [Solved]: Solution (1/2) — From Residuals to $W$

| | XLB | XLE | XLF | XLI | XLK | XLP | XLU | XLV | XLY |
|---|---|---|---|---|---|---|---|---|---|
| $\hat{\alpha}$ (% a year) | $-0.1$ | 2.4 | $-2.3$ | 0.5 | 0.3 | 1.7 | 3.4 | 1.8 | 0.4 |
| NW s.e. (% a year) | 2.6 | 4.2 | 2.7 | 1.6 | 2.5 | 1.8 | 2.5 | 1.6 | 1.9 |
| $\hat{\beta}$ | 1.05 | 0.92 | 1.07 | 1.05 | 1.28 | 0.45 | 0.40 | 0.67 | 1.07 |

1. No alpha is significant alone; the largest $|t|$ is XLU's, $3.39/2.53 = 1.34$
2. $\hat{\mathrm{E}}$: $331 \times 9$; $\hat{\boldsymbol{\Sigma}}$: $9 \times 9$; monthly alphas (%): $-0.007, 0.197, -0.191, 0.043, 0.027, 0.142, 0.282, 0.153, 0.036$; $\hat{\boldsymbol{\alpha}}^\top \hat{\boldsymbol{\Sigma}}^{-1} \hat{\boldsymbol{\alpha}} = 0.0153$
3. $\hat{\mu}_m = 0.659\%$, $\hat{\sigma}_m = 4.51\%$, $SR_m = 0.146$; $W = \frac{321}{9} \times \frac{0.0153}{1+0.146^2} = 35.67 \times 0.0150 = 0.53$, $p = 0.85$
4. Same 331 months, 25 size $\times$ B/M portfolios: GRS $= 2.91$, $p = 9 \times 10^{-6}$

- Interpretation: the non-rejection comes from the test assets, not from the sample; sectors mix small and big, value and growth
- Not rejecting is not accepting the CAPM: choose test assets that spread the characteristics suspected to be priced

# B2 [Solved]: Solution (2/2) — Chart and Interpretation




- $W = 0.53$ lies in the body of $F(9, 321)$, far below the 5% critical value 1.91: do not reject $H_0$: all $\alpha_i = 0$
- Sharpe-ratio form (A4): the sectors raise the monthly Sharpe ratio only from 0.146 to 0.191 (annual 0.51 → 0.66)

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# C1 [Proposed]: Critique an AI Answer — the Answer

- **Prompt** sent to an AI assistant: "`P` holds cleaned month-end adjusted prices of 10 BVB blue chips, one column per stock. Build a 12-month momentum long–short portfolio, top 3 minus bottom 3 with equal weights, and report its annual mean and volatility"
- **Answer**: the code below and the conclusion

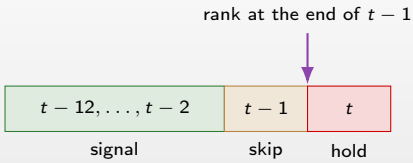
```
R   = np.log(P).diff()                          # monthly returns
mom = P.pct_change(12)                          # past 12-month return
ls  = {}
for t in R.index[13:]:
    s = mom.loc[t].dropna().sort_values()       # rank the stocks
    ls[t] = R.loc[t, s.index[-3:]].mean() - R.loc[t, s.index[:3]].mean()
ls = pd.Series(ls)
print(12 * ls.mean(), np.sqrt(12) * ls.std())  # annual mean, volatility
```

- **Conclusion of the AI**: "Momentum earns about 48% a year with a volatility of 16%. This agrees with Fama & French (2012), who document momentum on the Bucharest Stock Exchange."

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# C1 [Proposed]: Critique an AI Answer — Task

- **Strategy intended by the student** (the contract): 12–1 momentum (What You Need (11/21)); a stock is eligible if the signal and the price at the end of $t-1$ exist; long the top 3, short the bottom 3, equal weights, simple returns of month $t$
- **Data**: 10 blue chips (TLV, SNP, BRD, TGN, FP, SNG, SNN, EL, DIGI, TEL; H2O excluded: listed in July 2023), month-end, December 2015 – August 2026; the FP return of September 2023 ($|r| > 50\%$, a corporate event not adjusted in the series) is treated as missing
- **Tasks**
  1. Audit the timing of the signal, the aggregation of returns across stocks and the citation; list every error you find.
  2. For each error, say how you would detect it: a test, a published number or the paper itself.
  3. Write the corrected code and report the annual mean, volatility and number of months.
  4. Log each error in the format of `AI_ERRORS.md`: request, answer, error, how found, correction.
- **Report**: the list of errors, the corrected code, the three numbers and the `AI_ERRORS.md` entries




- **Interpretation**
  - does "about 48% a year" survive?
  - at which hurdle (1.96 or 3) is the corrected premium significant?
- AI code usually runs without an error message: the mistakes are in the finance, not in the syntax
- Model: B1; Notebook: section C1

# A3 [Solved]: Vasicek as a Posterior Mean — Task

- **Question**: why does the Vasicek rule shrink a noisy beta towards the cross-sectional mean?
- **Data**: 9 sector ETFs, monthly excess returns on SPY, January 1999 – December 2012; OLS betas with Newey–West errors
  - XLK: $\hat{\beta} = 1.415$, se $= 0.139$; cross-section: $\bar{\beta} = 0.949$, $\mathrm{Var}_{cs}(\hat{\beta}) = 0.1159$, $\overline{\mathrm{se}^2} = 0.0077$; XLK beta estimated on 2013–2026: 1.172
- **Assumptions**: prior $\beta_i \sim N(\bar{\beta}, \sigma_\beta^2)$; likelihood $\hat{\beta}_i \mid \beta_i \sim N(\beta_i, \mathrm{se}_i^2)$ with $\mathrm{se}_i$ known
- **Tasks**
  1. Multiply prior and likelihood, complete the square in $\beta$ and state the posterior distribution: its mean $\beta_i^V$ and its variance.
  2. Use the law of total variance to show that $\mathrm{Var}_{cs}(\hat{\beta}) = \sigma_\beta^2 + \mathbb{E}[\mathrm{se}^2]$, and write the empirical-Bayes estimator of $\sigma_\beta^2$.
  3. Compute $\sigma_\beta^2$, the weight $w$, $\beta^V$ and the posterior standard deviation for XLK, and the beta adjusted by the practitioners' rule $0.33 + 0.67\hat{\beta}$ (the Blume beta, What You Need (12/21)).
  4. Compare the absolute errors of the raw, Vasicek and Blume betas against the later estimate 1.172.
- **Report**: the posterior, the numbers of step 3 and the three errors
- Notebook: section A3

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# A3 [Solved]: Solution (1/3) — The Posterior

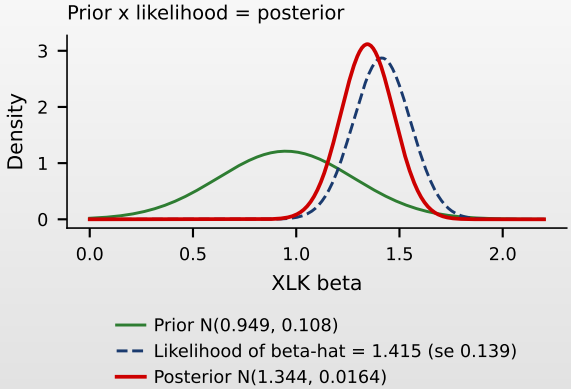
1. Posterior $\propto \exp\{-\frac{(\beta-\bar{\beta})^2}{2\sigma_{\beta}^2}\}\exp\{-\frac{(\hat{\beta}-\beta)^2}{2\,\mathrm{se}^2}\}$; the exponent is

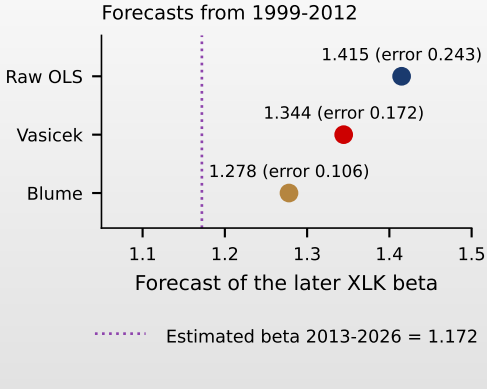
$$-\tfrac{1}{2}\Big[\beta^2\Big(\tfrac{1}{\sigma_{\beta}^2}+\tfrac{1}{\mathrm{se}^2}\Big)-2\beta\Big(\tfrac{\bar{\beta}}{\sigma_{\beta}^2}+\tfrac{\hat{\beta}}{\mathrm{se}^2}\Big)\Big]+\text{const}$$

   - Completing the square: a Normal posterior with variance $v=(1/\sigma_{\beta}^2+1/\,\mathrm{se}^2)^{-1}$
   - and mean $v(\bar{\beta}/\sigma_{\beta}^2+\hat{\beta}/\,\mathrm{se}^2)=w\hat{\beta}+(1-w)\bar{\beta}$, with $w=\sigma_{\beta}^2/(\sigma_{\beta}^2+\mathrm{se}^2)$

2. Law of total variance: $\mathrm{Var}(\hat{\beta})=\mathrm{Var}(\mathbb{E}[\hat{\beta}\mid\beta])+\mathbb{E}[\mathrm{Var}(\hat{\beta}\mid\beta)]=\sigma_{\beta}^2+\mathbb{E}[\mathrm{se}^2]$
   - Empirical Bayes: $\hat{\sigma}_{\beta}^2=\max\{\mathrm{Var}_{cs}(\hat{\beta})-\overline{\mathrm{se}^2},\,0\}$
   - Using $\mathrm{Var}_{cs}(\hat{\beta})$ itself counts sampling noise as true dispersion and shrinks too little

# A3 [Solved]: Solution (2/3) — Chart




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# A3 [Solved]: Solution (3/3) — Numbers and Forecasts

3. $\hat{\sigma}^2_\beta = 0.1159 - 0.0077 = 0.1082$; $\text{se}^2 = 0.0193$; $w = 0.849$; $\beta^V = 0.849 \times 1.415 + 0.151 \times 0.949 = 1.344$; posterior variance 0.0164 (s.d. 0.128); Blume 1.278
4. Errors against 1.172: raw 0.243, Vasicek 0.172, Blume 0.106

- XLK's beta is precise, so Vasicek shrinks it little; Blume moves every beta one third of the way to 1
- The later beta is itself an estimate; A5 (extension) asks why regression to the mean exceeds sampling noise

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# B3 [Solved]: PCA and the Number of Factors — Task

- **Questions**
  - how many common factors drive the daily returns of the US sectors?
  - what are these factors?
- **Data**: 11 sector ETFs, daily log returns on common days, 20 June 2018 – 18 September 2026 ($T = 2073$); SPY held out for comparison
- **Tasks**
  1. Standardise each column: Z is $T \times 11$; compute $\mathsf{C} = \mathsf{Z}^\top \mathsf{Z}/T$, its eigenvalues $\ell_1 \geq \cdots \geq \ell_{11}$ and unit-length eigenvectors, signed so that their loadings sum to a positive number.
  2. Report the shares $\ell_k/11$ of PC1–PC3 and the loadings of PC1 and PC2; build the scores $\mathsf{Z}\gamma_1$ ($\gamma_1$: the PC1 eigenvector) and correlate them with SPY.
  3. Estimate the number of factors, with $k_{\max} = 5$, by Kaiser, Bai–Ng ($IC_{p2}$) and Ahn–Horenstein (eigenvalue ratio), as defined in What You Need (14/21).
- **Report**: shares, loadings, the correlation with SPY, the three estimates of $k$ and a chart
- **Interpretation**: PC2 opposes utilities and staples to technology
  - would PC2 earn a premium?
  - what would you need to test a PC2 premium?
- Notebook: section B3

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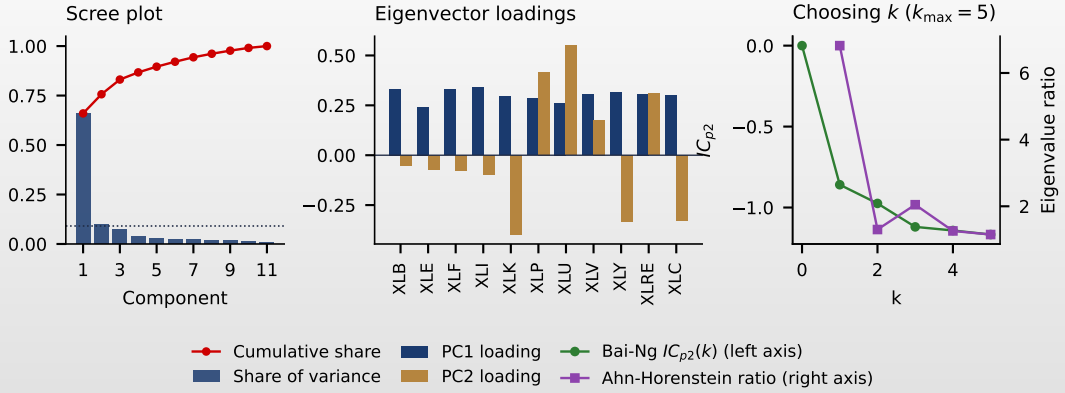
## B3 [Solved]: Solution (1/3) — Components

1. Eigenvalues $7.26, 1.06, 0.82, 0.40, 0.32, \ldots$ (they sum to 11)
2. Shares: PC1 66.0%, PC2 9.7%, PC3 7.4%; the first three explain 83.1%
   - PC1 loadings all positive (0.24–0.34): an almost equal-weighted portfolio; corr(PC1 scores, SPY) $= 0.954$: PC1 is strongly market-related, but not identical to SPY
   - PC2: XLU $+0.55$, XLP $+0.42$, XLRE $+0.31$ against XLK $-0.40$, XLY $-0.33$: defensive versus growth sectors
   - The eigenvectors are portfolio weights on standardised returns, not betas; a return beta on PC1 needs a regression of each sector on the scores
3. Number of factors: Kaiser 2, Bai–Ng 5, Ahn–Horenstein 1 (computation: Solution 3/3)
   - the estimators disagree: with $N = 11$ the large-$N$ asymptotics do not apply; report the range, not a single $k$

⊡ Interpretation: PCA explains covariance, not average returns, so PC2 need not be a priced factor

- test: Fama–MacBeth with PC2 betas on many assets, or a long–short portfolio sorted on PC2 loadings, its mean tested with Newey–West errors

# B3 [Solved]: Solution (2/3) — Chart




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# B3 [Solved]: Solution (3/3) — How Many Factors?

- Kaiser: $\hat{k} = 2$
  - two eigenvalues exceed 1: 7.26 and 1.06
- Bai–Ng: $\hat{k} = 5 = k_{\max}$
  - the penalty is $\frac{N+T}{NT} \ln 11 = 0.219$ per factor
  - $IC_{p2}(k)$ for $k = 0, \ldots, 5$: $0, -0.86, -0.97, -1.12, -1.14, -1.17$; it falls all the way to $k_{\max}$
- Ahn–Horenstein: $\hat{k} = 1$
  - the ratios $\ell_k / \ell_{k+1}$ for $k = 1, \ldots, 5$: $6.82, 1.30, 2.05, 1.26, 1.15$; the maximum is at $k = 1$
- To report: one dominant, market-related factor and at most a weak second one; Bai–Ng is not informative with $N = 11$

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# B4 [Proposed]: CAPM Betas of BVB Blue Chips — Task

- **Questions**
  - how sensitive are the BVB blue chips to the BET index?
  - how do stale prices bias daily betas?
- **Data**: 11 blue chips (adjusted closing prices) and the BET index; each stock aligned separately with BET on common days; daily log returns from 6 January 2015 (or the listing date) to 18 September 2026; a constant $R_f$ cancels
- **Tasks**
  1. Form the lead and the lag of BET on the full calendar of each pair, then drop days with $|r_t| > 0.5$ for the stock or BET; report the dropped days.
  2. Estimate each beta by OLS on BET, with Newey–West errors (lag rule) and $R^2$.
  3. Compute empirical-Bayes Vasicek betas (A3), the 11 stocks as the cross-section.
  4. Estimate the Dimson betas, with BET at $t-1$, $t$ and $t+1$ (What You Need (15/21)).
- **Report**: a table of the four betas and a chart with 95% intervals

**Cleaning example: DIGI around 23 May 2017**

| Date | $r_{\text{DIGI}}$ | $\text{BET}_{t-1}$ | $\text{BET}_t$ | $\text{BET}_{t+1}$ | keep |
|---|---|---|---|---|---|
| 19 May | 1.78% | −0.31% | 0.65% | 0.22% | yes |
| 22 May | 2.73% | 0.65% | 0.22% | 0.05% | yes |
| 23 May | −658.7% | 0.22% | 0.05% | 0.48% | no |
| 24 May | −2.97% | 0.05% | 0.48% | 0.48% | yes |
| 25 May | 1.74% | 0.48% | 0.48% | 0.66% | yes |

- A scale error in the first records (the closes of 17–22 May are about 725 times larger) creates a log return of −6.59; the day is dropped
- The lead and the lag of the kept days still use BET on 23 May: form them before dropping
- **Interpretation**
  - which stocks does the Dimson correction move most?
  - why those stocks?
- Model: B1, A3; Notebook: section B4

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# Extension Packs: What Each One Needs

| Pack | Exercises | Prerequisites |
|---|---|---|
| 1. Pricing restrictions | A2 Extended [Solved], A4, A5 | A2, A3, B2 |
| 2. Two-pass inference | A6 [Solved], A7, B5, B6, B7 | B1, B2, lecture Section 6 |
| 3. Model comparison | B8 | B1, lecture Section 7 |
| 4. Multiple testing on real alphas | B9 | A1, B1 |
| 5. Latent factors | A8, B10 | B3 |
| 6. Daily factor ETFs | B11 | B1 |
| 7. BVB factor premia | C2 | B4, C1 |
| 8. Replication project | C3 | A3, A1, lecture case study |

- Each pack has its own notebook section and starter code; do them in order within a pack
- Useful for the team project: C2 and C3 are possible project starts

# A2 Extended [Solved]: Other Efficient Portfolios — Task

- **Question**: does the beta pricing equation need the market portfolio, or only an efficient benchmark?
- **Inputs**: as in A2; two benchmarks
  - a CML portfolio, $a\mathrm{w}_m$ in risky assets ($a > 0$) and $1 - a$ in the risk-free asset
  - a frontier portfolio of risky assets only (gross returns R, means m): the first-order condition of $\min \mathrm{w}^\top \boldsymbol{\Sigma} \mathrm{w}$ s.t. $\mathrm{w}^\top \mathrm{m} = m^*$, $\mathrm{w}^\top 1 = 1$ gives $\boldsymbol{\Sigma}\mathrm{w} = \kappa_1 \mathrm{m} + \kappa_2 1$, with $\kappa_1$, $\kappa_2$ the Lagrange multipliers of the two constraints
- **Tasks**
  1. Show that the CML portfolio has $\beta_{i,p} = \beta_i / a$ and $\mu_p = a\mu_m$, so $\mu_i = \beta_{i,p}\mu_p$ again.
  2. From the first-order condition, show that $m_i - \gamma = \beta_{i,p}(m_p - \gamma)$ with $\gamma = -\kappa_2/\kappa_1$, the zero-beta rate.
  3. State what this implies for a test of the CAPM with a market proxy.
- **Report**: the two proofs and one sentence on what a CAPM test checks
- Model: A2

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## A2 Extended [Solved]: Solution

1. The excess return of the CML portfolio is $aR_m^e$: $\mathrm{Cov}(R_i^e, aR_m^e) = a\,\mathrm{Cov}(R_i^e, R_m^e)$ and $\mathrm{Var}(aR_m^e) = a^2\sigma_m^2$, so $\beta_{i,p} = \beta_i/a$; with $\mu_p = a\mu_m$, $\beta_{i,p}\mu_p = \beta_i\mu_m = \mu_i$
2. $\mathrm{Cov}(\mathrm{R}, R_p) = \boldsymbol{\Sigma}\mathrm{w} = \kappa_1\mathrm{m} + \kappa_2 1$ and $\sigma_p^2 = \mathrm{w}^\top\boldsymbol{\Sigma}\mathrm{w} = \kappa_1 m_p + \kappa_2$
   - Hence $\kappa_1(\mathrm{m} - \gamma 1) = \mathrm{Cov}(\mathrm{R}, R_p)$ and $\kappa_1(m_p - \gamma) = \sigma_p^2$; divide: $\mathrm{m} - \gamma 1 = \boldsymbol{\beta}_p(m_p - \gamma)$
   - $\gamma$ is the mean of every portfolio with zero beta on $p$: the zero-beta CAPM of Black (1972)
3. Any mean–variance efficient benchmark prices all assets exactly by beta; a rejection means that the proxy is not efficient for the test assets, which is a statement about the proxy as much as about the model (Roll's critique, Roll (1977))

# A4 [Proposed]: GRS Equals a Sharpe-Ratio Gain — Task

- **Question**: why does GRS measure how much the test assets improve the market's Sharpe ratio?
- **Notation**: $Y_t = (f_t, \mathrm{R}_t^{e\top})^\top$, $(N+1) \times 1$; ML moments $\boldsymbol{\mu}_Y = (\mu_f, \boldsymbol{\mu}_R^\top)^\top$ and $\mathrm{V} = \begin{pmatrix} \sigma_f^2 & \boldsymbol{\sigma}_{Rf}^\top \\ \boldsymbol{\sigma}_{Rf} & \boldsymbol{\Sigma}_R \end{pmatrix}$, all divided by $T$; $\boldsymbol{\beta} = \boldsymbol{\sigma}_{Rf}/\sigma_f^2$, $\boldsymbol{\alpha} = \boldsymbol{\mu}_R - \boldsymbol{\beta}\mu_f$, $\boldsymbol{\Sigma} = \boldsymbol{\Sigma}_R - \boldsymbol{\beta}\boldsymbol{\beta}^\top \sigma_f^2$
- **Supplied identity** (block inverse with the Schur complement $\boldsymbol{\Sigma}$ of $\sigma_f^2$)

$$\mathrm{V}^{-1} = \begin{pmatrix} 1/\sigma_f^2 + \boldsymbol{\beta}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\beta} & -\boldsymbol{\beta}^\top \boldsymbol{\Sigma}^{-1} \\ -\boldsymbol{\Sigma}^{-1}\boldsymbol{\beta} & \boldsymbol{\Sigma}^{-1} \end{pmatrix}$$

- **Tasks**
  1. Expand $SR^2(f, \mathrm{R}) = \boldsymbol{\mu}_Y^\top \mathrm{V}^{-1} \boldsymbol{\mu}_Y$ into four terms and show that it equals $\mu_f^2/\sigma_f^2 + \boldsymbol{\alpha}^\top \boldsymbol{\Sigma}^{-1}\boldsymbol{\alpha}$.
  2. Write the GRS statistic $W$ as a function of $SR^2(f)$ and $SR^2(f, \mathrm{R})$.
  3. The 25 size $\times$ B/M portfolios: $T = 757$, $N = 25$, monthly $SR(f) = 0.134$, $SR(f, \mathrm{R}) = 0.405$; compute $W$ and decide at 5% (critical value of $F(25, 731)$: 1.52).
  4. Compare the observed gain with the gain expected under $H_0$, $N(1 + SR^2(f))/(T - N - 3)$.
- **Report**: the algebra, $W$, the decision and one sentence
- Model: A2, B2; Notebook: section A4

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# A5 [Proposed]: Blume's Slope as a Shrinkage Weight — Task

- **Question**: is the regression of later betas on earlier betas a Vasicek estimator?
- **Simplifying assumptions**: true betas constant, $\hat{\beta}_{i,k} = \beta_i + \eta_{i,k}$ in periods $k = 1, 2$; the noises are independent of $\beta_i$ and of each other, with a **common** variance $s^2$; $\beta_i$ has mean $\bar{\beta}$ and variance $\sigma^2_\beta$
- **Tasks**
  1. Derive, as $N \to \infty$, the limits of the OLS slope $b$ and intercept $a$ of $\hat{\beta}_2$ on $\hat{\beta}_1$.
  2. Show that $a + b\hat{\beta}_1$ is then the Vasicek beta with $w = b$.
  3. With A3's $\sigma^2_\beta = 0.10822$, $s^2 = 0.00773$, $\bar{\beta} = 0.9495$, compute the implied $b$ and $a$; compare them with the fitted 0.7242 and 0.2304 and plot both lines.
  4. The fitted slope has standard error 0.1137: compute $(0.7242 - b)/0.1137$ as an illustration and list the explanations that remain possible.
- **Report**: the two limits, the implied $b$ and $a$, the chart of the two lines and the list of explanations
- Model: A3; Notebook: section A5

| | $\hat{\beta}_1$ 1999–2012 | se$_1$ | $\hat{\beta}_2$ 2013–2026 |
|---|---|---|---|
| XLB | 1.174 | 0.083 | 1.026 |
| XLE | 0.904 | 0.089 | 1.051 |
| XLF | 1.217 | 0.120 | 1.029 |
| XLI | 1.122 | 0.057 | 1.078 |
| XLK | 1.415 | 0.139 | 1.172 |
| XLP | 0.451 | 0.064 | 0.578 |
| XLU | 0.463 | 0.086 | 0.455 |
| XLV | 0.716 | 0.047 | 0.705 |
| XLY | 1.083 | 0.063 | 1.169 |

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# A6 [Solved]: Errors in Variables in Pass 2 — Task

- **Question**: how much does pass 2 attenuate a premium because the betas are estimated?
- **Setting**: one factor, $R_{it} = a_i + \beta_i f_t + \varepsilon_{it}$; $\varepsilon_{it}$ i.i.d. over $t$ and $i$, variance $\sigma_\varepsilon^2$, independent of $f$ and of the $\beta_i$; the $\beta_i$ i.i.d. across assets with variance $\mathrm{Var}_{cs}(\beta)$
- **Three premia**: $\lambda$, the true price of risk; $\mu_f = \mathbb{E}[f_t]$; $\lambda_T = \lambda + \bar{f} - \mu_f$, the realised premium in the sample. Pricing restriction: $a_i = \lambda_0 + \beta_i(\lambda - \mu_f)$
- Work conditionally on the factor path $f_1, \ldots, f_T$; pass 2: the OLS regression of $\bar{R}_i$ on $(1, \hat{\beta}_i)$, $\hat{\beta}_i = \beta_i + \eta_i$, $\eta_i = \sum_t (f_t - \bar{f})\varepsilon_{it} / \sum_t (f_t - \bar{f})^2$
- **Tasks**
  1. Show that $\mathrm{Var}(\eta_i \mid f) = \sigma_\varepsilon^2/(T\hat{\sigma}_f^2)$ and that $\mathrm{Cov}(\eta_i, \bar{\varepsilon}_i \mid f) = 0$.
  2. Show that, for $N \to \infty$ and $T$ fixed, the pass-2 slope converges to $\lambda_T \ \mathrm{Var}_{cs}(\beta)/[\mathrm{Var}_{cs}(\beta) + \sigma_\varepsilon^2/(T\hat{\sigma}_f^2)]$.
  3. Compute the attenuation factor for the two panels below, with $\mathrm{Var}_{cs}(\beta) = \mathrm{Var}_{cs}(\hat{\beta}) - \bar{\sigma}_\varepsilon^2/(T\hat{\sigma}_f^2)$, and explain what the Shanken correction adjusts.
- **Report**: the two proofs, the two attenuation factors and one sentence on the Shanken correction; Notebook: section A6

| Panel (CAPM) | $T$ | $\mathrm{Var}_{cs}(\hat{\beta})$ | $\bar{\sigma}_\varepsilon^2$ (%² per month) | $\hat{\sigma}_f^2$ (%² per month) |
|---|---|---|---|---|
| 25 size × B/M, 1963–2026 | 757 | 0.01905 | 9.33 | 19.87 |
| 9 sector ETFs, 1999–2026 | 331 | 0.09247 | 13.45 | 20.36 |

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# A6 [Solved]: Solution (1/2) — Derivation

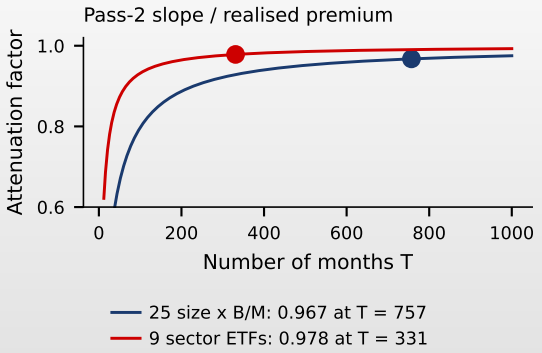
1. Given $f$, $\eta_i$ is a linear combination of the $\varepsilon_{it}$ with weights $(f_t - \bar{f}) / \sum_s (f_s - \bar{f})^2$
   - $\mathrm{Var}(\eta_i \mid f) = \sigma_\varepsilon^2 \sum_t (f_t - \bar{f})^2 / [\sum_s (f_s - \bar{f})^2]^2 = \sigma_\varepsilon^2 / (T \hat{\sigma}_f^2)$
   - $\mathrm{Cov}(\bar{\varepsilon}_i, \eta_i \mid f) = \sigma_\varepsilon^2 \frac{1}{T} \sum_t (f_t - \bar{f}) / \sum_s (f_s - \bar{f})^2 = 0$, because $\sum_t (f_t - \bar{f}) = 0$
2. With the restriction, $\bar{R}_i = \lambda_0 + \lambda_T \beta_i + \bar{\varepsilon}_i$: the realised factor mean $\bar{f}$ is common to all assets and does not average out as $N \to \infty$
   - Given $f$, $(\beta_i, \eta_i, \bar{\varepsilon}_i)$ are i.i.d. across $i$: $\widehat{\mathrm{Cov}}_{cs}(\bar{R}, \hat{\beta}) \to \lambda_T \, \mathrm{Var}_{cs}(\beta)$ and $\widehat{\mathrm{Var}}_{cs}(\hat{\beta}) \to \mathrm{Var}_{cs}(\beta) + \sigma_\varepsilon^2 / (T \hat{\sigma}_f^2)$
   - The ratio is the stated limit: attenuation of $\lambda_T$ towards zero; $\lambda_T - \lambda = \bar{f} - \mu_f$ vanishes only as $T \to \infty$

# A6 [Solved]: Solution (2/2) — Numbers

3. 25 portfolios: noise $9.33/(757 \times 19.87) = 0.00062$; $\text{Var}_{cs}(\beta) = 0.01905 - 0.00062 = 0.01843$; factor $0.01843/0.01905 = 0.967$

   9 sectors: noise $13.45/(331 \times 20.36) = 0.00200$; $\text{Var}_{cs}(\beta) = 0.09048$; factor 0.978

- The bias is only 2–3% here: diversified portfolios have small residual variance and the sample is long; single stocks have a much larger $\sigma_\varepsilon^2$, so the attenuation grows
- Shanken (1992) corrects the standard error of $\hat{\lambda}$ for the noise in $\hat{\beta}$ (a variance correction); it does not remove the finite-$T$ bias of the point estimate




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# A7 [Proposed]: A Useless Factor in Pass 2 — Task

- **Question**: why can a factor unrelated to returns look priced in pass 2?
- **Assumptions**: $N$ fixed, $T \to \infty$; $R_t$ i.i.d. over time with means $\mu_i$, nonsingular covariance $\mathbf{\Sigma}$, finite fourth moments; a factor $g_t$, i.i.d., mean 0, variance $\sigma_g^2$, independent of all returns, so $\beta_{i,g} = 0$; in the proof $g$ is the only factor in pass 2
- **Tasks**
  1. Show that $\sqrt{T}\,\hat{\beta}_{i,g} \to N(0, \sigma_i^2/\sigma_g^2)$: $\hat{\beta}_{i,g}$ is of order $T^{-1/2}$ (numerator and denominator separately).
  2. Write $\hat{\lambda}_g = \widehat{\mathrm{Cov}}_{cs}(\bar{R}_i, \hat{\beta}_{i,g}) / \widehat{\mathrm{Var}}_{cs}(\hat{\beta}_{i,g})$ with $z = \sqrt{T}\,\hat{\boldsymbol{\beta}}_g$; show $\hat{\lambda}_g$ is of order $\sqrt{T}$ when the $\mu_i$ differ.
  3. Case of equal means: using the joint limit of $(\sqrt{T}(\bar{R} - \boldsymbol{\mu}), z)$, show that $\hat{\lambda}_g$ converges to a non-degenerate random variable, not 0.
  4. The lecture simulation adds $g$ to the CAPM on the 25 portfolios (2000 replications): $\lambda_g = 0$ is rejected in 59.6% of the replications with Fama–MacBeth errors and in 26.6% with Shanken errors, while the first-pass Wald test of $\beta_{i,g} = 0$ rejects in 6.7%; explain these rates and state which first-pass test protects against them.
- **Report**: the three asymptotic results and the explanation of the rejection rates
- Model: A6; Notebook: section A7

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# B5 [Proposed]: Fama–MacBeth on ETFs — Task

- **Question**: are there market, size and value premia in 16 US ETFs over 2013–2026?
- **Data**: 9 sector ETFs and 7 factor ETFs (MTUM momentum, VLUE value, QUAL quality, USMV minimum volatility, IWM Russell 2000, IWD Russell 1000 Value, IWF Russell 1000 Growth), monthly excess returns, August 2013 – July 2026 ($T = 156$): a $156 \times 16$ panel; factors Mkt-RF, SMB of the FF3 model (three-factor data set of the Kenneth French Data Library), HML: $156 \times 3$
- **Tasks**
  1. Pass 1: estimate the full-sample regression of each ETF on the three factors; report $\hat{B}$ ($16 \times 3$).
  2. Pass 2: for each month, estimate the OLS regression of the 16 returns on $X = [1, \hat{B}]$; show the computation of one month in full (fitted against actual), then average the 156 slopes; annualise by multiplying by 12.
  3. Compute the Fama–MacBeth, Newey–West (on $\hat{\lambda}_t$, 6 lags) and Shanken standard errors (What You Need (16/21); no factor term for the intercept); as an extension, compute the Kan–Robotti–Shanken errors with the notebook helper `two_pass_gmm`.
  4. Compare each $\hat{\lambda}$ with the time-series mean of its factor in a chart and test $\lambda = 0$ at 5%.
- **Report**: $\hat{B}$, the month computed in full, the table of premia with their standard errors, the chart and the test decisions
- **Interpretation**: is a non-significant premium here evidence that size and value are not priced?
- Model: B1, B2 and lecture Section 6; Notebook: section B5

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# B6 [Proposed]: A Lewellen–Nagel–Shanken Check — Task

- **Question**: does FF3's high cross-sectional $R^2$ survive broader test assets and a GLS metric (Lewellen, Nagel & Shanken (2010))?
- **Data**: 25 size × B/M and 30 industry portfolios (Kenneth French Data Library), monthly excess returns, July 1963 – July 2026 ($T = 757$); FF3 factors
- **Definitions**: $\mu$, X, $\hat{V}$, $R^2_{OLS}$ and $R^2_{GLS}$ as in What You Need (18/21)
- **Tasks**
  1. Baseline: FF3 on the 25 portfolios; plot actual against fitted means; report both $R^2$.
  2. Add the 30 industries and repeat; also impose $\lambda = \mathbb{E}[f]$ with no intercept and report $R^2_{OLS}$ with the same denominator.
  3. Uncertainty: resample months i.i.d. (500 draws), re-estimate both passes and $\hat{V}$ in each draw; report 95% intervals.
  4. Extension: repeat the analysis for the CAPM and FF5 on both sets of assets.
- **Report**: the table of the two $R^2$ with their intervals, on both sets of assets, and the chart
- **Interpretation**: does the FF3 $R^2$ survive once industries are added?
- Model: B2, lecture Section 6; Notebook: section B6

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# B7 [Proposed]: GRS without Normality — Task

- **Question**: is the rejection of the CAPM on the 25 portfolios an artefact of assuming residuals with the Normal distribution?
- **Data**: 25 size × B/M portfolios, CAPM with Mkt-RF, monthly, July 1963 – July 2026 ($T = 757$, $N = 25$)
- **Tasks**
  1. Diagnostics: estimate the 25 regressions; compute the Pearson kurtosis of each residual series (What You Need (9/21)) and its average; draw a Q–Q plot.
  2. Exact test: compute GRS with the ML $\hat{\boldsymbol{\Sigma}}$ and its $p$-value from $F(25, 731)$.
  3. Null bootstrap: resample whole rows of residuals (keeping their cross-correlation), rebuild $\mathrm{R}_t^* = \hat{\boldsymbol{\beta}} f_t + \boldsymbol{\varepsilon}_t^*$ (so $\alpha = 0$), refit and recompute GRS 2000 times; report $p = (1 + \#\{W^* \geq W\})/2001$ and the 95% bootstrap critical value.
  4. Extension: compute the robust Wald test with the moments $(1, f_t)^\top \otimes \boldsymbol{\varepsilon}_t$, parameters ordered $(\boldsymbol{\alpha}, \boldsymbol{\beta})$; $W = \hat{\boldsymbol{\alpha}}^\top \hat{\mathrm{V}}_\alpha^{-1} \hat{\boldsymbol{\alpha}} \to \chi^2_{25}$ with the sandwich $\mathrm{D}^{-1}\hat{\mathrm{S}}\mathrm{D}^{-1}/T$, $\mathrm{D} = (\mathrm{Z}^\top \mathrm{Z}/T) \otimes \mathrm{I}_{25}$, $\mathrm{Z} = [1, \mathrm{f}]$, and $\hat{\mathrm{S}}$ White or Newey–West (6 lags).
  5. Explain why GRS cannot be computed for the 100 size × B/M portfolios over August 2018 – July 2026 ($T = 96$).
- **Report**: the average kurtosis, the Q–Q plot, $W$ with the $p$-values of the three methods and the bootstrap critical value
- **Interpretation**: does the departure of the residuals from the Normal distribution change the decision of the GRS test?
- Model: B2; Notebook: section B7

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# B8 [Proposed]: Comparing Models by Sharpe Ratios — Task

- **Questions**
  - which factor model spans the highest Sharpe ratio?
  - is the difference significant (Barillas & Shanken (2018); Barillas, Kan, Robotti & Shanken (2020))?
- **Data**: monthly Mkt-RF, SMB, HML, RMW, CMA, MOM (Kenneth French Data Library), July 1963 – July 2026 ($T = 757$) and January 2000 – July 2026 ($T = 319$)
- **Membership**: CAPM (Mkt-RF); FF3 and Carhart use the three-factor SMB, FF5 and FF5 + MOM the five-factor one; covariances divide by $T - 1$
- **Tasks**
  1. Compute the maximum squared Sharpe ratio $SR^2$ of each model (What You Need (19/21)), annualise ($\times 12$) and report $\sqrt{SR^2}$.
  2. Nested models, FF5 + MOM against FF5: regress MOM on the five FF5 factors and test the intercept (spanning regression, Newey–West).
  3. Non-nested models, FF5 against Carhart: apply a stationary bootstrap (Politis & Romano (1994)) with a mean block of 6 months and 2000 draws, with the same indices for both models; report the 95% interval of the $SR^2$ difference.
  4. Extension: compute the spanning $t$ of each of the six factors on the other five, in both periods.
- **Report**: the Sharpe ratios in both periods, the spanning $t$ and the bootstrap interval
- **Interpretation**
  - why can the percentile interval of a nested difference never contain 0?
  - which factors add no significant alpha after 2000?
- Model: B1; Notebook: section B8

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# B9 [Proposed]: Multiple Testing of 25 Alphas — Task

- **Question**: how many of the 25 size × B/M portfolios have a genuine FF3 alpha?
- **Data**: 25 value-weighted size × B/M portfolios (the "25 Portfolios Formed on Size and Book-to-Market" data set of the Kenneth French Data Library) and the FF3 factors (Fama & French (1993)), monthly excess returns, July 1963 – July 2026 ($T = 757$)
- **Tasks**
  1. For one portfolio (small-growth), show the full computation: FF3 regression, Newey–West $t$ of $\alpha$ (6 lags), two-sided $p = 2(1 - \Phi(|t|))$, $\Phi$: the distribution function of the standard Normal distribution.
  2. Repeat for all 25; draw a $5 \times 5$ heatmap of the annual alphas with their $t$-statistics.
  3. Sort the $p$-values and apply the A1 worksheet: naive, Bonferroni, Holm, Benjamini–Hochberg; extension: Benjamini–Yekutieli.
  4. Count the portfolios with $|t| > 3$ (Harvey, Liu & Zhu (2016)), a heuristic hurdle, not a procedure with a guaranteed error rate.
- **Report**: the small-growth computation, the heatmap, the number of discoveries of each procedure and the number of portfolios with $|t| > 3$
- **Interpretation**: the 25 tests are strongly correlated
  - which methods remain valid?
  - which test uses the correlation directly?
- Model: A1; Notebook: section B9

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# A8 [Proposed]: Equicorrelated Assets — Task

- **Question**: how does the share of the first component depend on the average correlation and on $N$?
- **Stage 1**, $N = 3$, $\rho = 0.5$: $C = \begin{pmatrix} 1 & 0.5 & 0.5 \\ 0.5 & 1 & 0.5 \\ 0.5 & 0.5 & 1 \end{pmatrix}$
  1. Verify that $v_1 = (1,1,1)/\sqrt{3}$, $v_2 = (1,-1,0)/\sqrt{2}$, $v_3 = (1,1,-2)/\sqrt{6}$ are orthonormal eigenvectors and find the eigenvalues.
  2. Compute the PC1 share $\ell_1/3$.
- **Stage 2**, $N$ assets, $C = (1-\rho)I + \rho 1 1^\top$
  3. Show that $\ell_1 = 1 + (N-1)\rho$ and that all other eigenvalues equal $1-\rho$; find the limit of the PC1 share as $N \to \infty$.
  4. Invert the share formula to get the equicorrelation-equivalent $\rho$ of a PC1 share $s$; apply it to the BVB blue chips ($N = 10$, $s = 0.425$, B10) and the US sectors ($N = 11$, $s = 0.660$, B3).
  5. Extension: for a general C with average off-diagonal correlation $\bar{\rho}$, show that $1 + (N-1)\bar{\rho} \le \ell_1$.
- **Report**: the eigenvalues, the formula of the PC1 share with its limit and the two values of $\rho$
- Model: B3; Notebook: section A8

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# B10 [Proposed]: PCA of BVB Blue Chips — Task

- **Question**: is the first factor of the Romanian blue chips as strong as in the US sectors?
- **Data**: 10 blue chips (TLV, SNP, BRD, TGN, FP, SNG, SNN, EL, DIGI, TEL; H2O excluded, listed in 2023) and BET, aligned on common days from 7 June 2017 to 18 September 2026; daily log returns; drop days with any $|r_t| > 0.5$ and report how many; BET is held out of the PCA
- **Tasks**
  1. Apply the B3 recipe: standardise ($T \times 10$), eigenvalues, variance shares, PC1 loadings.
  2. Correlate the PC1 scores with the BET returns.
  3. Estimate the number of factors by Kaiser, Bai–Ng and Ahn–Horenstein, with $k_{\max} = 5$.
  4. Compare with the US sectors on the same period (from 20 June 2018): PC1 shares and scree plots side by side.
- **Report**: the number of days dropped, the shares, the PC1 loadings, the correlation with BET, the three estimates of $k$ and the scree plots
- **Interpretation**
  - what does a weaker first factor imply for diversification within the BVB?
  - how much of the gap comes from comparing stocks with sector portfolios?
- Model: B3; Notebook: section B10

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# B11 [Proposed]: Do Factor ETFs Have Alpha? — Task

- **Question**: do MTUM and QUAL earn an alpha beyond their six-factor exposures?
- **Data**: adjusted prices of MTUM (18 Apr 2013 – 31 Jul 2026) and QUAL (18 Jul 2013 – 31 Jul 2026); daily FF5 (Fama & French (2015)), MOM (Carhart (1997)) and RF, same end date
- **Calendar rule**: keep an ETF return only if the previous ETF price dates from the immediately preceding trading day of the factor calendar
  - Example: MTUM has prices on 18 and 23 April 2013 but none on 19 and 22 April; the return 18 → 23 April covers three factor days and is dropped
- **Tasks**
  1. Build daily excess returns with the calendar rule; report $n$ before and after alignment.
  2. Estimate $R^e_t = \alpha + b\,\mathrm{MKT}_t + s\,\mathrm{SMB}_t + h\,\mathrm{HML}_t + r\,\mathrm{RMW}_t + c\,\mathrm{CMA}_t + m\,\mathrm{MOM}_t + \varepsilon_t$; annualise $\alpha$ by multiplying by 252.
  3. Compute the Newey–West standard error (10 lags) and $t$ of $\alpha$; apply a moving-block bootstrap of the pairs, with blocks of 20 days and 1000 draws, and report the 95% interval.
  4. Plot the two intervals for each ETF.
- **Report**: $n$ before and after alignment, the six loadings, $\alpha$ with its s.e. and $t$, the two intervals and the chart
- **Interpretation**: why must a 3-day ETF return never be regressed on a 1-day factor return?
- Model: B1; Notebook: section B11

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# C2 [Proposed]: Do Factor Premia Exist on the BVB? — Setting

- **Question**: do momentum (Jegadeesh & Titman (1993)) and low beta earn premia among BVB blue chips?
- **Data**: month-end adjusted prices of the 10 blue chips of C1 and BET, December 2015 – August 2026 (September 2026 incomplete); monthly simple returns; the FP return of September 2023 ($|r| > 50\%$, a corporate event not adjusted in the series) is treated as missing for signals and holding returns
- **Rules fixed in advance**
  - signal and beta use information up to the end of $t-1$ (timeline of C1)
  - at least 6 eligible stocks; ties broken by ticker order
  - a missing holding return is dropped from the leg mean

**Ranking for August 2026**

| Stock | Signal | Leg |
|---|---|---|
| EL | 129.9% | long |
| TGN | 127.3% | long |
| DIGI | 113.2% | long |
| SNG | 88.9% | |
| BRD | 87.8% | |
| SNN | 75.5% | |
| TEL | 66.1% | |
| FP | 51.0% | short |
| TLV | 48.5% | short |
| SNP | 46.6% | short |

- Signal: the compounded return of August 2025 – June 2026, known at the end of July 2026

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# C2 [Proposed]: Do Factor Premia Exist on the BVB? — Tasks

- **Tasks**
  1. Build 12–1 momentum (C1 contract): long the top 3, short the bottom 3, equal weights.
  2. Build the low-beta portfolio: estimate 36-month rolling betas on BET (at least 18 months, ending at $t-1$); long the 3 lowest, short the 3 highest; an unscaled spread, not the beta-neutral BAB of Frazzini & Pedersen (2014).
  3. Report periods, $T$, annual mean, volatility and Newey–West $t$ (3 lags); compare $t$ with 1.96 and 3; plot the cumulative spreads.
  4. Discuss survivorship (today's blue chips chosen with hindsight), short-selling constraints, costs and liquidity.
  5. Extension: identify public sources of book equity and shares outstanding, needed for a BVB value factor.
- **Deliverable**: a Quantlet folder in the team repository and one slide with the conclusion and its main limitation
- Model: B4, C1; Notebook: Part C

# C3 [Proposed]: Do US Factors Replicate? A Staged Project (1/2)

- **Context**: the lecture case study, Jensen, Kelly & Pedersen (2023): 82.4% of US factors keep a significant CAPM alpha under a hierarchical empirical-Bayes model, against 75.6% with Benjamini–Yekutieli and 35% in Hou, Xue & Zhang (2020) (Figure 1)
- **Data**: value-weighted univariate sorts of the Kenneth French Data Library, July 1963 – July 2026, and Mkt-RF from the three-factor data set; long–short extremes on 15 characteristics
  - extreme 30%: size, book-to-market, operating profitability, investment, earnings/price, cash flow/price, dividend/price; extreme 20%: accruals, net share issues, market beta, variance, residual variance; extreme 10%: prior returns of months 12–2 (the 12–1 momentum of What You Need (11/21)), 1–0 and 60–13
- **Stage 1, construction and frequentist rates**
  1. Build the 15 factors, long the side that earned more in the original paper; cite that paper for each sign.
  2. Regress each factor on Mkt-RF, rescale it to a monthly residual volatility of $10\%/\sqrt{12}$ (Section II.A of the paper), report $\hat{\alpha}$, $t$ and the OLS replication rate ($t \geq 1.96$).
  3. Apply Benjamini–Yekutieli at 5% and report the critical $|t|$ and the rate.

# C3 [Proposed]: Do US Factors Replicate? A Staged Project (2/2)

- **Stage 2, themes and the hierarchical model**
  4. Build the Ward dendrogram on the distance $1-$ correlation of the CAPM residuals; compare it with the paper's themes (`Cluster Labels.csv`, authors' code).
  5. Build M ($15 \times J$), the theme-membership matrix; estimate $\tau_c, \tau_w$ by maximum likelihood of $\hat{\boldsymbol{\alpha}} \sim N(0, \mathrm{M}\mathrm{M}^\top \tau_c^2 + \mathrm{I}\tau_w^2 + \hat{\boldsymbol{\Sigma}}/T)$, report a boundary estimate $\tau_c = 0$ if it occurs; compute the posterior means and variances (Proposition 4) and the empirical-Bayes rate (posterior $z \geq 1.96$).
- **Stage 3, sensitivity**: compute the rate with a single theme and as a function of $\tau_c$, then the Bayesian FDR (Proposition 5)
- **Stage 4, post-sample decay**: estimate the alphas within the original sample years (from July 1963) and after them, with at least 60 months in each; regress the post-sample on the in-sample alphas by feasible GLS with the inverse factor covariance; compare the slope with 0.26 (Figure 8)
- **Report**: a factor table (alpha, $t$, theme, posterior mean, $z$), the rates next to Figure 1, $\hat{\tau}$ next to Table I, the dendrogram and the slope; keep the paper's numbers and the course numbers apart
- Model: A3, A1 and the lecture case study

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